



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECJ2203: Software Engineering

System Documentation (SD)

KADA's Cooperative System

Version X

8/12/2024

Faculty of Computing

Prepared by: team KACIP

Revision Page

a. **Overview**

b. **Target Audience**

stakeholders

c. **Project Team Members**

Member Name	Role	Task	Status
Muhammad Naim Bin Abdullah			
Muhammad Mukhritz Al iman bin Mohd Raffi			
Syarifah Dania Binti Syed Abu Bakar			
Yasmin Batrisya Binti Zahiruddin			

d. **Version Control History**

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Team leader 1 (Full Name)	Completed Chapter X, Section...	dd/mm/yyyy
...			

Note: The title page should show the latest version. For your first submission (PR2-Chap.1 and 2), it should be 1.0, then 2.0 for PR3 and 3.0 for PR4. Primary author(s) could be the team leaders who should compile all materials for the purpose of SD submission. [remove this note in red texts and the examples in the table above]

Note:

This System Documentation (SD) template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998), Software Design Descriptions (SDD) (IEEE Std. 1016-1998 1), and Software Test Documentation (IEEE Std. 829-2008) that are simplified and customized to meet the need of SECJ2203 course at School of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly.

Table of Contents

1	Introduction		1
	1.1	Purpose	1
	1.2	Scope	?
	1.3	Definitions, Acronyms and Abbreviations	?
	1.4	References	?
	1.5	Overview	?
2	Specific Requirements		?
	2.1	Persona	?
		2.1.1 Persona 1 (User type 1/Actor 1) 2.1.2 Persona 2 (User type 2/Actor 2) 2.1.3 Persona 3 (User type 3/Actor 3) 2.1.n Persona n (User type n/Actor n)	
	2.2	System Features 2.2.1 Use Case Diagram 2.2.2 Domain Model: Class Diagram 2.2.3 State Diagram	
	2.3	Launch Phase	
	2.4	User Story Details	
		2.4.1 US001: User Story <User Story 1> User Story description of US001 Activity Diagram of US001 System Sequence Diagram of US001 2.4.2 US002: User Story <User Story 2> User Story description of US002 Activity Diagram of US002	

		System Sequence Diagram of US002 2.4.3 US002: User Story <User Story 3> User Story description of US003 Activity Diagram of US003 System Sequence Diagram of US003 2.4.n USxxn: User Story <User Story n> User Story description of US00n Activity Diagram of US00n System Sequence Diagram of US00n	
	2.5	Performance and Other Requirements (Non Functional)	
	2.6	Design Constraints	

1. Introduction

1.1 Purpose

A Software Requirements Specification (SRS) serves as a guide for the development team and clients to ensure an agreement of the system's objectives, features, and limitations by precisely defining and documenting the functional and non-functional requirements of a software system.

In the case of the proposed KADA Registration and Loan Application System, this SRS document defines the requirements for digitizing KADA's existing manual processes for membership registration and loan applications. The document will serve as a reference and guidance for stakeholders, including developers, project managers and testers, outlining the project's scope, functional and non-functional requirements, and design considerations, providing a roadmap for the development and implementation phases.

1.2 Scope

The proposed system at KADA means to digitize and ease up the present manual procedures about membership registration and loan applications, respectively. This enhances efficiency in terms of minimizing paperwork, reducing error cases, and providing a practical platform for members, applicants, and KADA personnel. There are three roles that the system plays at the front-applicants, clerks, and ALK.

The system provides the facility for online membership and loan application forms that can be filled out anytime and sent without necessarily going to submit them physically, especially to those who are far away from KADA's headquarters. Immediately after successful submission, applicants get confirmation to reduce uncertainty about the status of their forms.

It facilitates clerks' work through one common platform that will enable them to handle membership and loan applications efficiently. One can access applicant details, verify information submitted, and change the status of an application with ease. The system enables clerks to generate various reports and track applications, enhancing productivity.

The application will be reviewed and approved by ALK members remotely via the system, which is a very significant improvement compared to the present system where their presence in meetings is absolutely necessary. The efficiency at which decision-making has been improved by providing access to ALK members from any location greatly reduces approval times.

In this respect, automated notifications, document storage, and real-time data access will make operations within KADA smoother and more effective. While the system focuses on registration and loan management, it is also designed with heavy security for the safety of sensitive personal and financial data. This is a modernization project that seeks to increase productivity among the staff in KADA, as well as improve the experience for applicants and members.

1.3 Definitions, Acronyms and Abbreviation (mukhriz)

- SD – System Documentation
- KADA - Lembaga Kemajuan Pertanian Kemubu
- ALK - Ahli Lembaga Koperasi
- SRS - System Requirement Specification
- Functional requirement - technical details on specification, data analysis and processing, and explain what a system must do
- Non-functional Requirement - criteria that will be used to judge the system operation

1.4 References (yas)

This subsection should:

- Provide a complete list of all documents referenced elsewhere in the SD;*
- Identify each document by title, report number (if applicable), date, and publishing organization;*
- Specify the sources from which the references can be obtained.*

Specify a complete list of references using a standardized reference format.

1.5 Overview

This SRS document is divided into sections that provide a comprehensive overview of the KADA system. It involves an overview of the system requirements, functional and non-functional specifications, user interfaces, data management rules, system constraints, and potential improvements. The system's goals and objectives are thoroughly examined in each section, with detailed descriptions, specifications, and requirements provided to assure understanding.

This document provides an overview for project stakeholders, including project managers and development teams, to better understand and visualize the KADA system's requirements and

specifications. This description of the planned KADA system promotes effective data storage, data accessibility and decision making.

2. Specific Requirements

*This section of SD serves as the SRS that should contain all the software requirements to a level of detail sufficient to enable designers to design a system to satisfy those requirements, and testers to test that the system satisfies those requirements. Throughout this section, every stated requirement should be externally perceivable by users, operators, or other external systems. These requirements should **include at a minimum a description of every input (stimulus) into the system, every output (response) from the system, and all functions performed by the system** in response to an input or in support of an output.*

2.1 Persona (dania)

Describes the intended users of the system and their characteristics, such as their technical expertise, knowledge, and physical abilities. This section is important because it provides the development team with a clear understanding of whom the system is designed for, and helps to ensure that the system is developed to meet the needs of its intended users.

2.1.1 Persona 1 (Applicant)

User Need

- Apply membership or loan from KADA without having to visit KADA's headquarters.
- Track loan status on the website

User Stories

- An applicant wants to track the status of their loan application so that they can know if it has been approved or needs further action. A status indicator is displayed in the dashboard and emails are sent for status updates.

2.1.2 Persona 2 (Clerk)

User Need

- Manage and verify applications efficiently
- Create user accounts for the ones whose membership applications are approved

User Stories

- A clerk wants to review membership applications to review and verify applicants' details

2.1.3 Persona n (Ahli Lembaga Pengarah (ALK))

User Need

- Review and approve loan applications.
- Access financial reports for further evaluation

User Stories

- ALK who wants to review and approve loan applications remotely to participate in decision making even when they are not in the office.

2.2 System Features

The KADA Cooperative System is a flexible software solution that works with all devices, including Android and iOS platforms. It improves the management of member registrations and loan applications by addressing issues about data storage and accessibility. The system's real-time cloud database ensures simple access and an improved user experience.

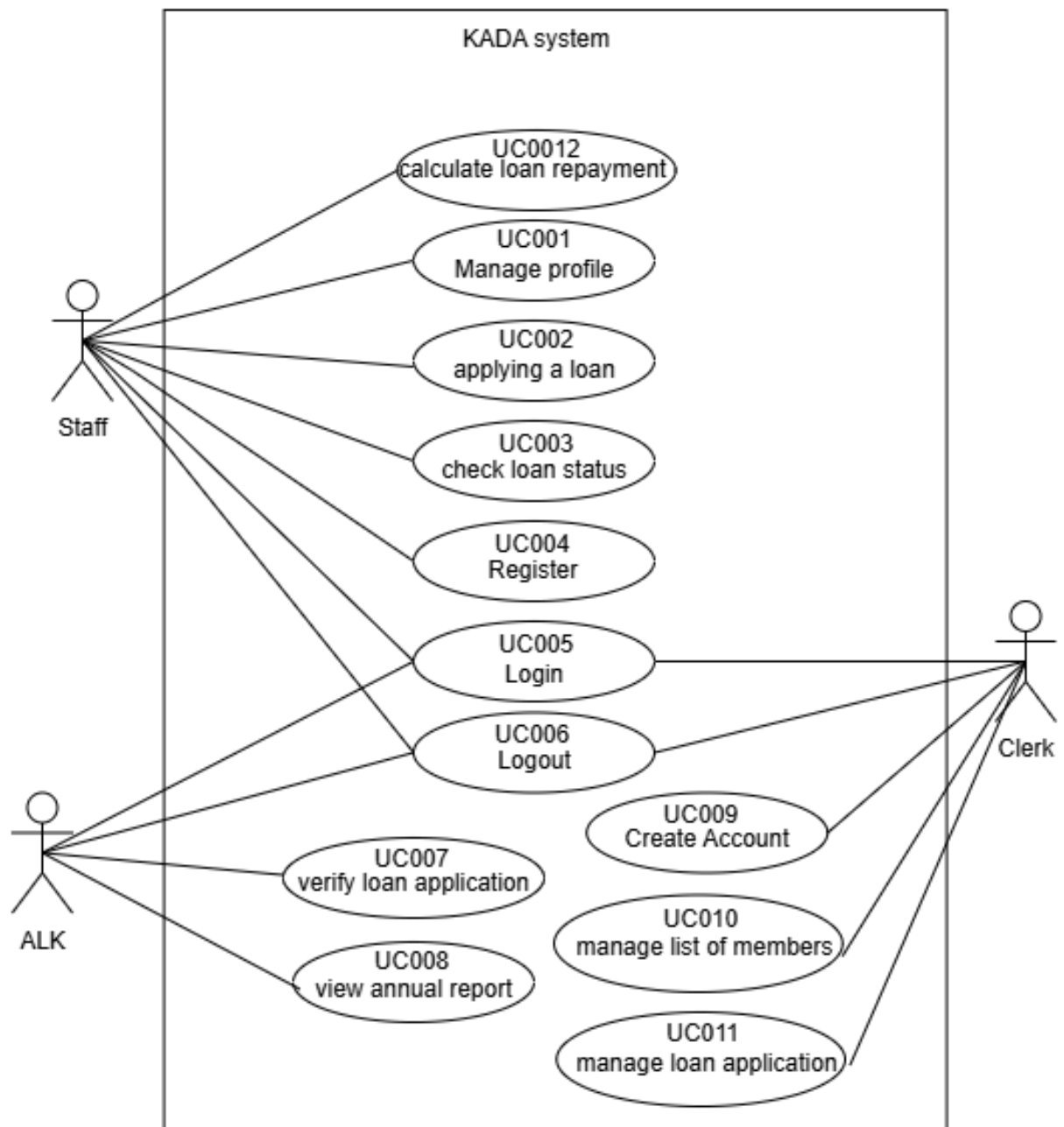


Figure 1.1: Use Case Diagram for KADA cooperative system

Table 1.1: Description of Module and Functions for KADA cooperative system

Use Case ID	Name	Description
UC001	Manage Profile	Allows an applicant to edit personal information.
UC002	Applying Loan	Allows an applicant to make a loan application.
UC003	Check Loan Status	Allows an applicant to check the status of their loan (approved or declined).
UC004	Register	Allows members to register as members of the KADA Cooperative.
UC005	Login	Allows users (applicants, clerks, and ALK) to access their respective interfaces.
UC006	Logout	Allows all users to log out of the system.
UC007	Verify Loan Application	Enables ALK to view loan application details and decide whether to approve or decline.
UC008	View Annual Report	Enables ALK to view the annual report of the KADA Cooperative, including profit or loss data.
UC009	Create an Account	Enables the clerk to create an account for an applicant after verifying their staff status.
UC010	Manage List of Members	Enables the clerk to delete, update, or view personal information of members.
UC011	Manage Loan Application	Enables the clerk to filter and review loan applications but not make approval decisions.
UC012	Calculate loan repayment	Allow member to calculate their own loan repayment

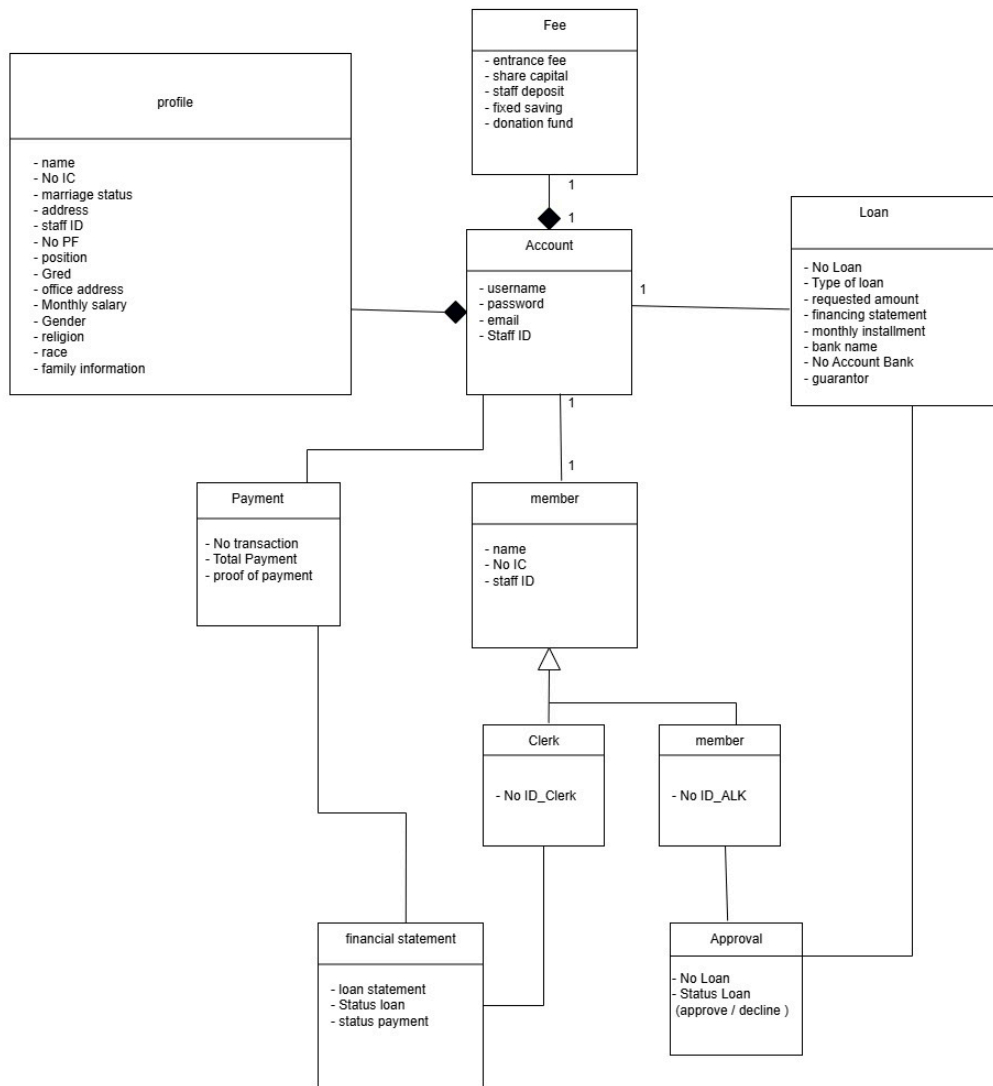
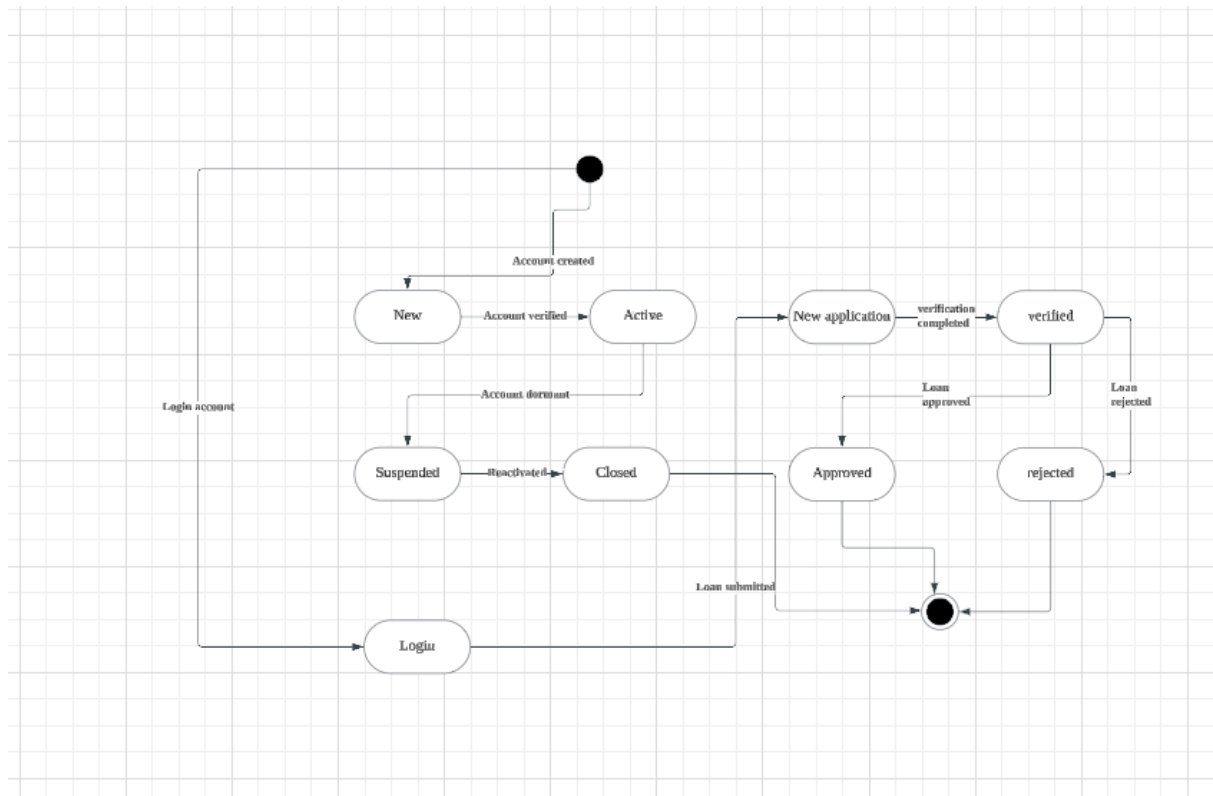


Figure 1.2: Domain Model for KADA cooperative system



2.3 Launch Phase

Product Backlog [describe more on product backlog];

Sprint	Team members assigned
UC001 - Manage Profile UC004 - login * UC005 - register * UC006 - Logout	Syarifah Dania Binti Syed Abu Bakar
UC002 - applying loan UC003 - check loan status UC012 - calculate loan payment	Yasmin Batrisyia Binti Zahiruddin
UC009 - create an account * UC010 - manage list of member UC011 - manage loan application	Muhammad Naim Bin Abdullah
UC007 - verify loan application	

UC008 - <i>view annual report</i>	Muhammad Mukhritz Al Iman bin Mohd Raffi
UC013 - <i>Payment</i>	

2.4 User Story Details

[Provide ID for each user story such as US001 and so on...]

Include User Story Description for each use case as in the example below. If there are any alternative flows, add the rows accordingly. Otherwise, just remove the rows. Ensure each User Story has its own unique ID and the name of the heading corresponds to the name of use case in the use case diagram (refer to Figure 2.1). Different scenarios of a user story require separate descriptions.]

2.4.1 US001: User Story <Manage Profile>

Table 2.1: User Story Description for <Manage Profile>

User story: <Manage Profile>
ID: UC001
User Story Description As a member of the cooperative I want to be able to manage my personal information in the system So that I can ensure my details are accurate and up to date
Flow of events: 1. The member logs into the system using their credentials (delegated username and password) 2. The member navigates to the "Profile" section in the dashboard 3. The member updates the desired fields (name, address, contact info) 4. The system validates the input and saves changes 5. A confirmation message is displayed to the customer ...
Alternative flow <i>n</i>: If a member attempts to update their email address with an invalid format 1. The system displays an error message indicating the email format is incorrect. 2. The member re-enters the email address in the correct format and submits again.
Acceptance Criteria Postcondition - The member's profile is updated in the system database Precondition - The member must be logged into the system with valid credentials Other conditions

- | |
|--|
| <ul style="list-style-type: none"> - The system must ensure that only the logged-in member can update their profile |
|--|

Exception flow:

If the system encounters a server error during the update:

1. The changes are not saved.
2. An error message is displayed to the customer

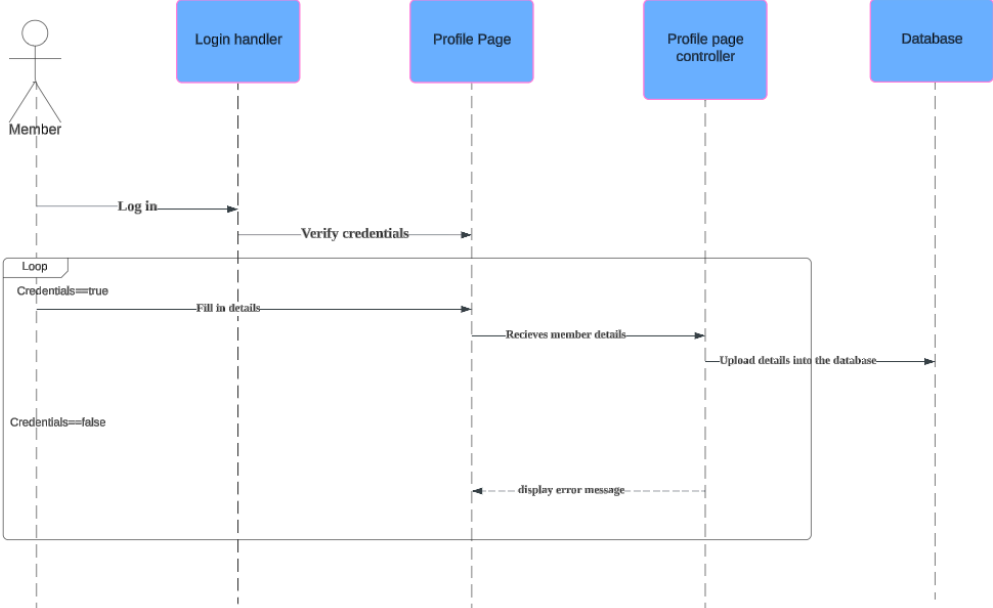


Figure 2.5: Sequence Diagram for <Manage Profile>

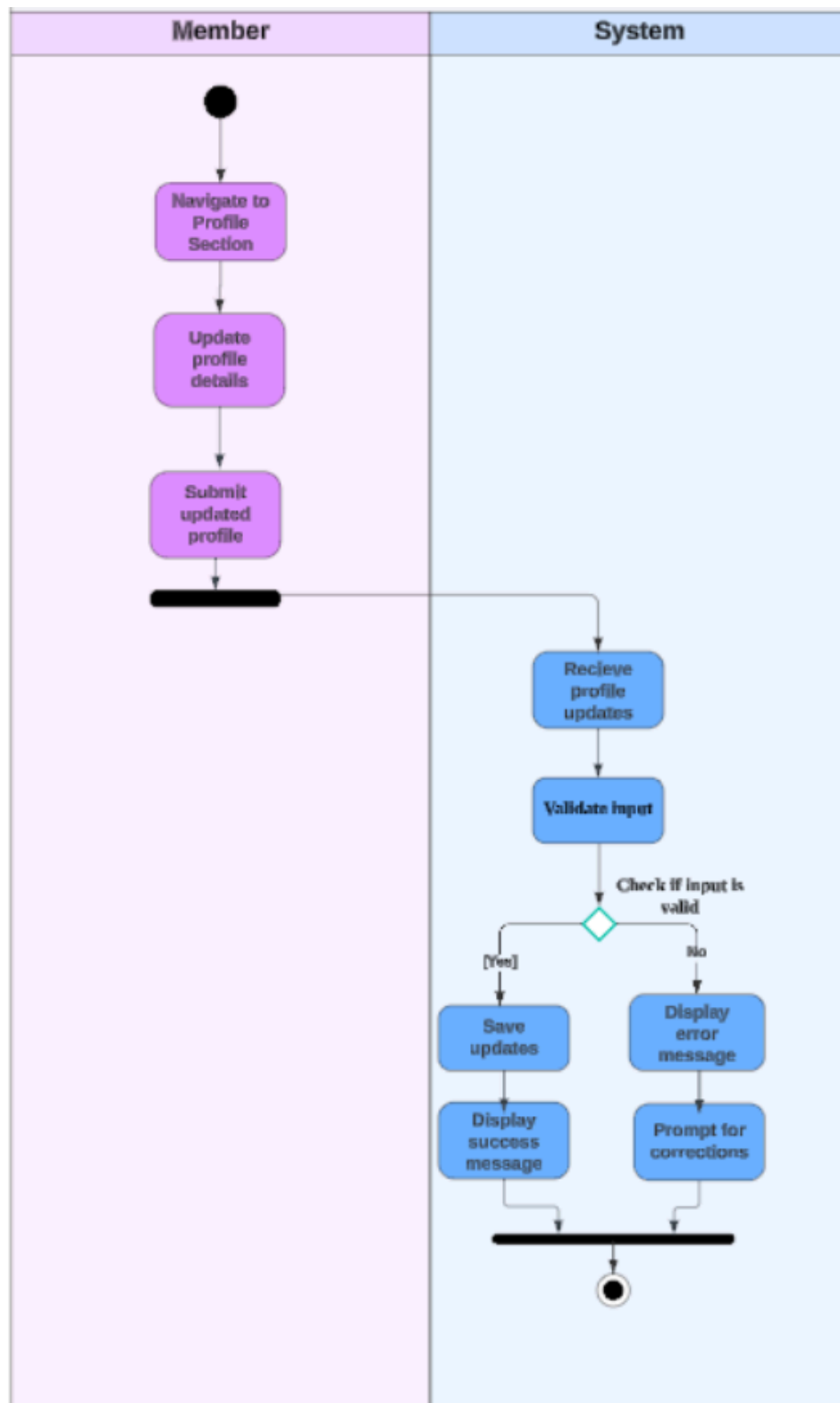


Figure 2.2: Activity Diagram for <<Manage Profile>>

2.4.2 US002: User Story <Applying Loan>

Table 2.2: User Story Description for <Applying Loan>

User story: <Applying Loan>
ID: US002
User Story Description As a cooperative member of KADA I want to be able to apply for a loan So that I can use the money for my needs without using up all my savings
Flow of events: 1. The member navigates to the loan application page 2. The member filled in required details 3. The member submit registration form 4. The system will receive the details and validate 5. If the details are filled in correctly, details is saved to the database 6. User will receive the successful application message ...
Alternative flow <i>n</i>: If the member provide an invalid information: 1. The system highlights the invalid or incomplete fields with error messages. 2. The customer corrects the information and resubmits.
Acceptance Criteria Postcondition - The member will get the notification says that the form has been submitted Precondition - User must be a membership of KADA
Exception flow: If the system fails during loan application submission - An error message is displayed: "Loan application failed, please try again later."

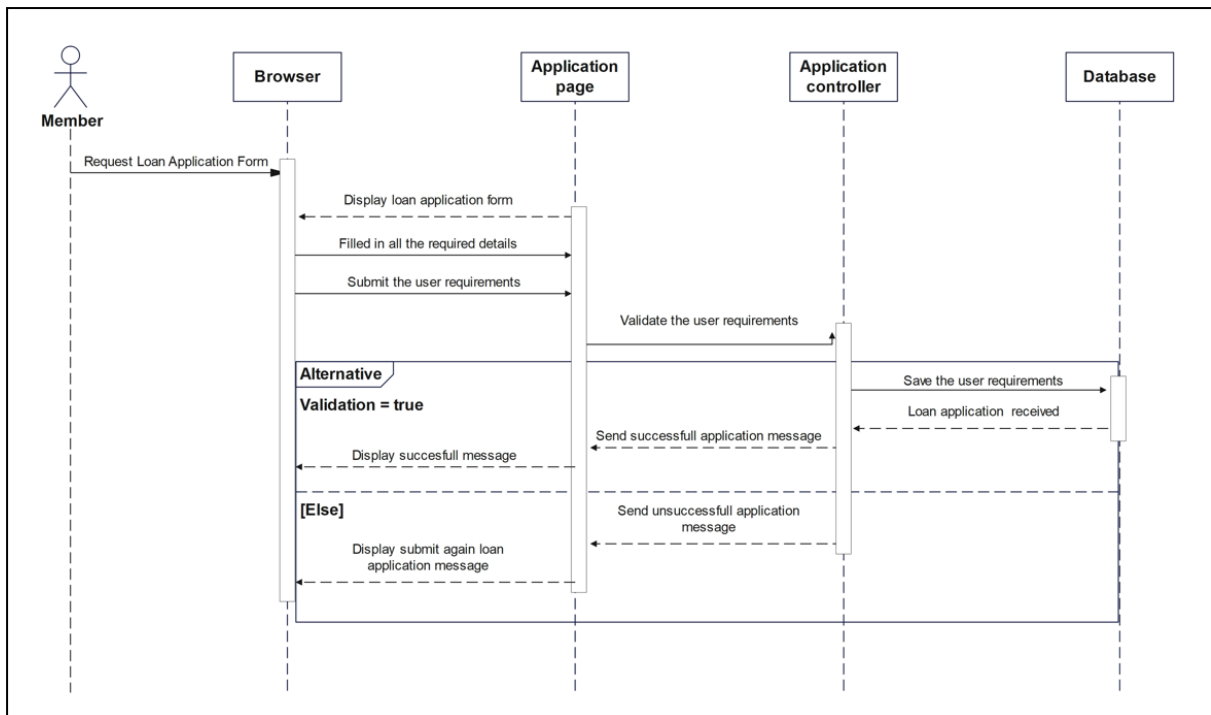


Figure 2.4.1: Sequence diagram for <Applying Loan>

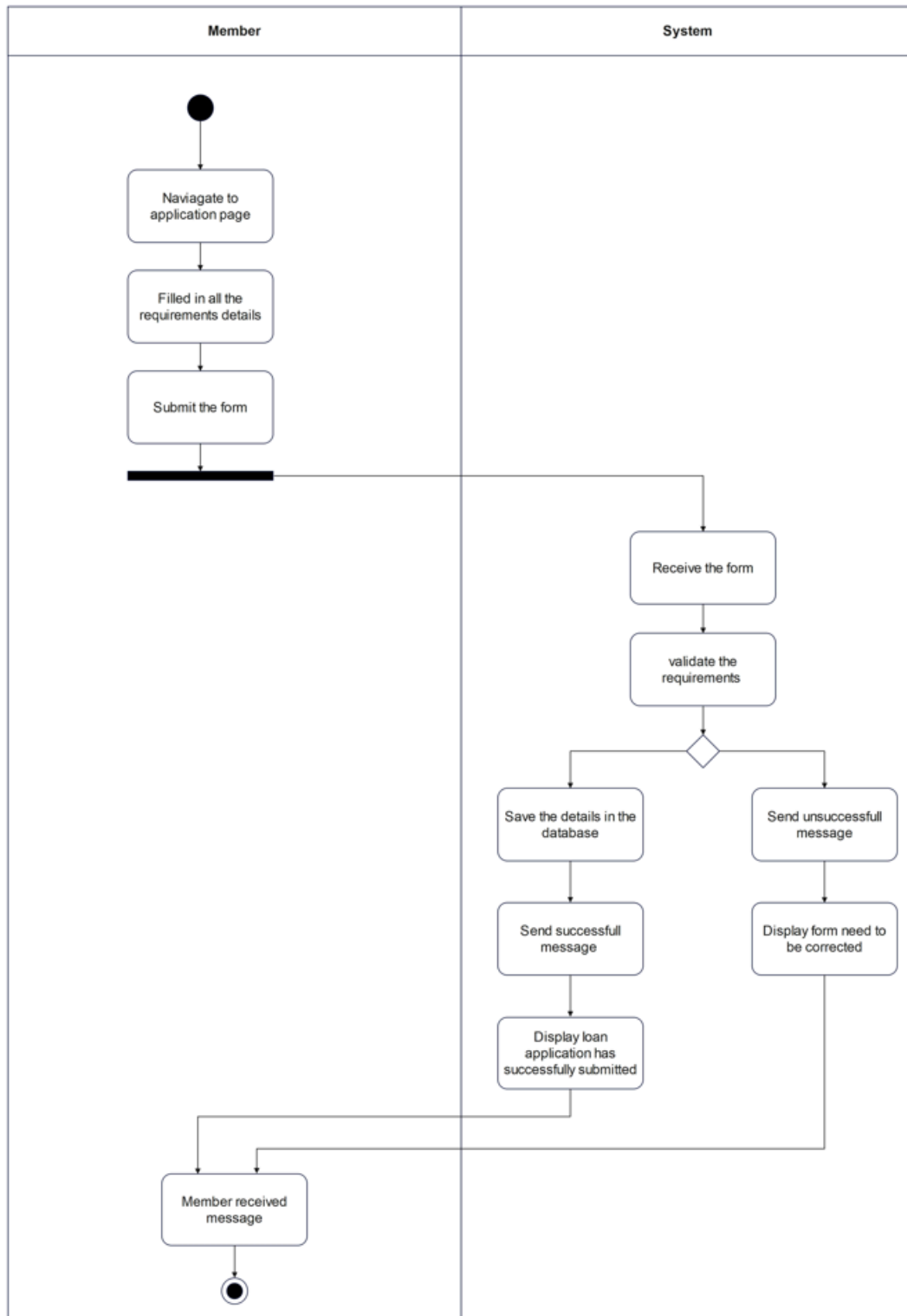


Figure 2.4.2 Activity Diagram for <Applying Loan>

2.4.3 US003: User Story <Check Loan Status>

Table 2.3: User Story Description for <Check Loan Status>

User story: <Check Loan Status>
ID: US003
User Story Description As a cooperative member of KADA I want to check my loan status after I have submitted the loan application So that I can acknowledge the current update regarding to the submission
Flow of events: 1. The member navigates to the profile page 2. Member click on the 'Semak Status' section 3. The system will appear the current loan application status to the member 4. The status will appear that the loan is successful, when the loan has been approved from ALK ...
Alternative flow <i>n</i>: If the loan application is unsuccessful: 1. Loan is not approved 2. Display that there is some error in the submission 3. The member is suggested to resubmit the form
Acceptance Criteria Postcondition - Loan application will appear the loan status is being processed, successful, or unsuccessful precondition - Member must log in to check the loan application status
Exception flow: If the status is not available due to a system error, the ALK member receives a notification to retry later

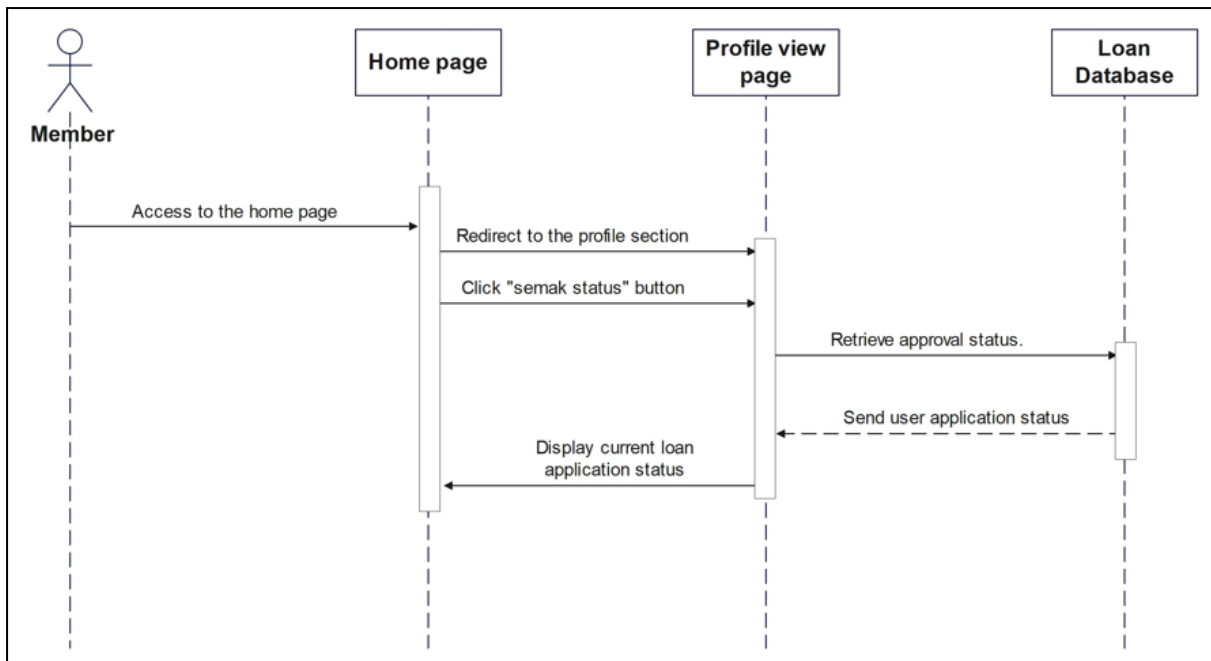


Figure 2.5.1: Sequence Diagram for <Check Loan Status>

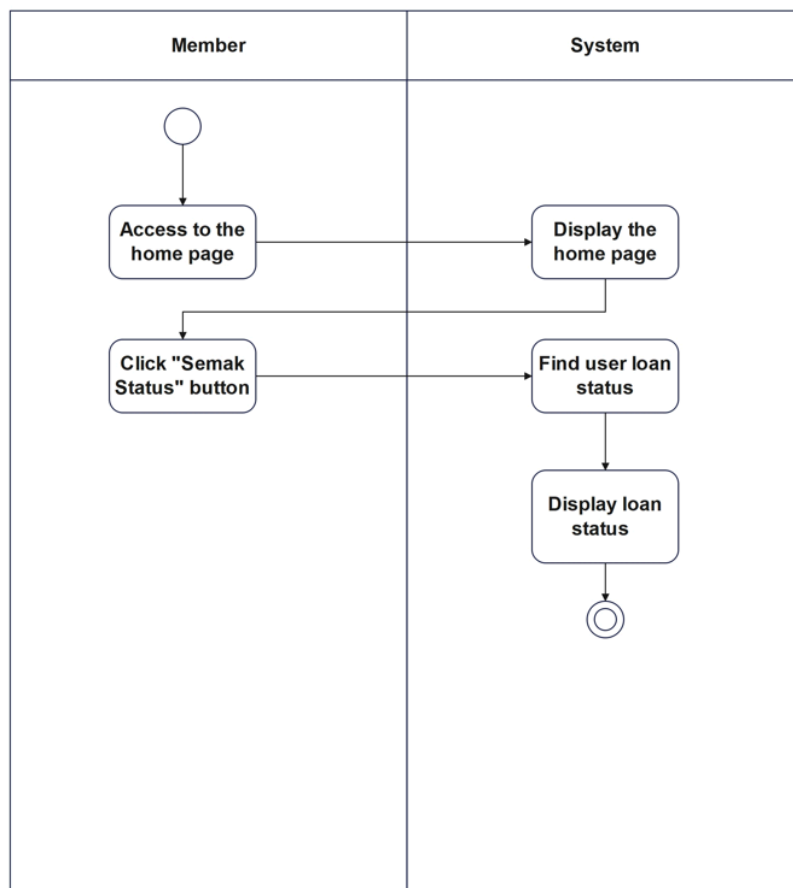
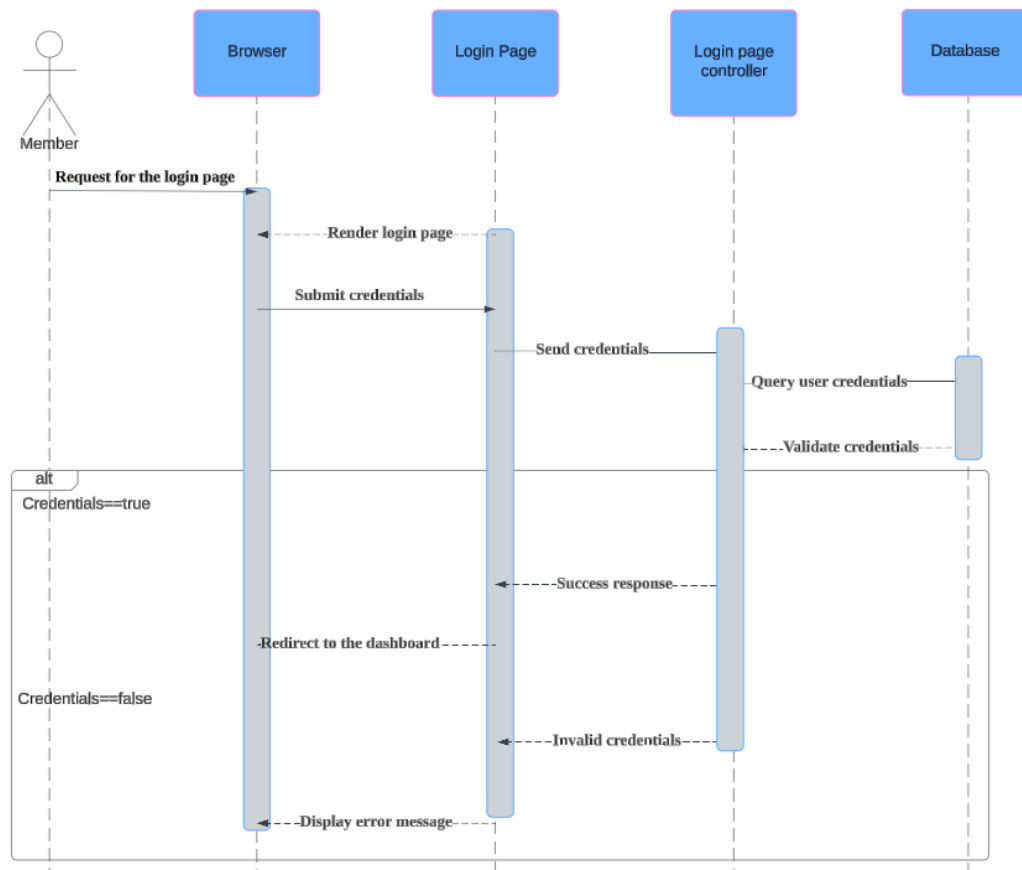


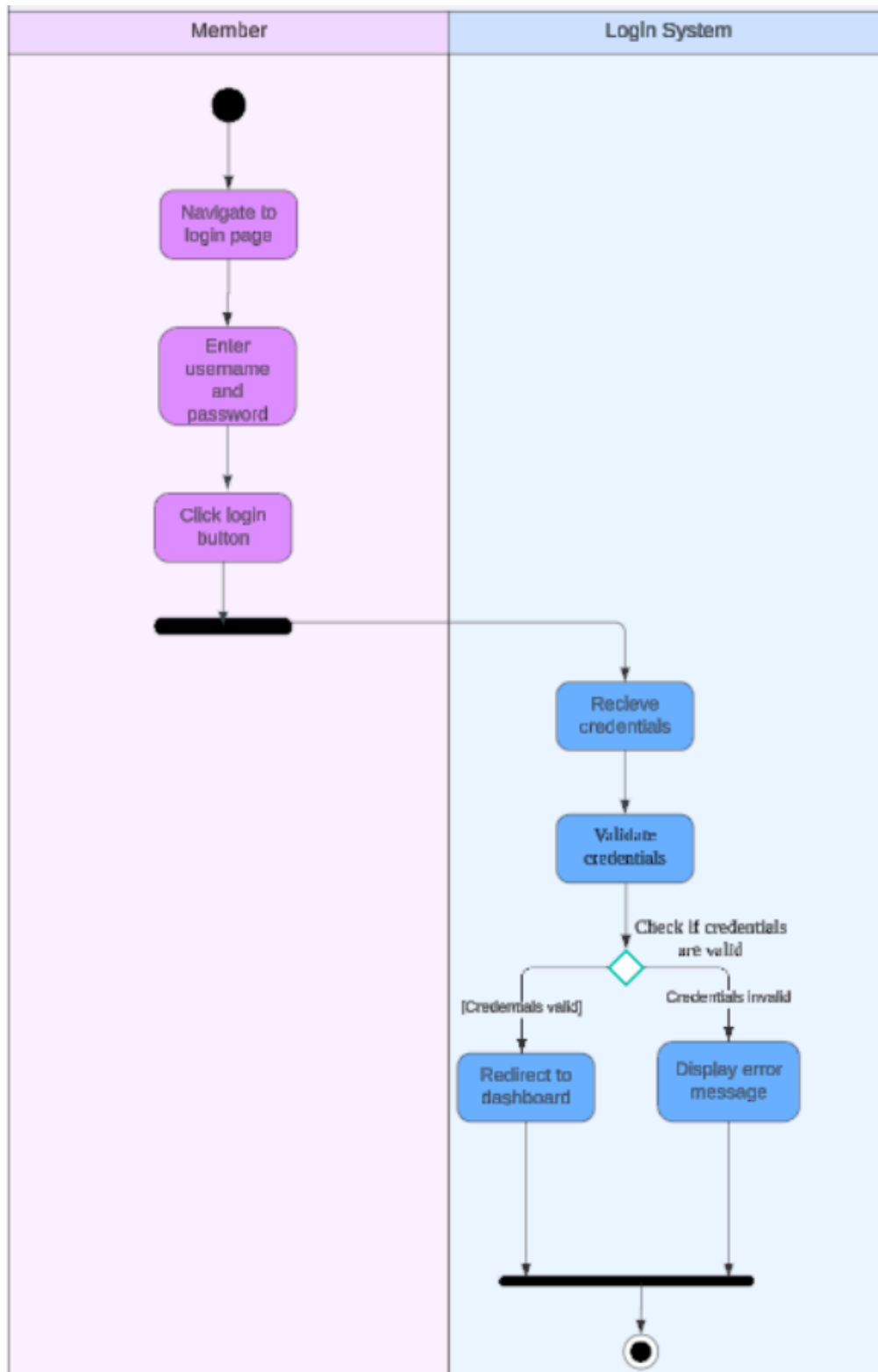
Figure 2.5.2: Activity Diagram for <check Loan Status>

2.4.4 US004: User Story <Login>

Table 2.4: User Story Description for <Login>

User story: <Login>
ID: US004
User Story Description As a member of the cooperative I want to be able to log into the system So that I can securely access my account and its features
Flow of events: 1. The member navigates to the login page 2. The member enters their username and password 3. The system validates their credentials 4. Upon successful validation, the member is redirected to their dashboard ...
Alternative flow <i>n</i>: If the member enters invalid credentials: 1. The system displays an error message "Invalid username or password." 2. The member is prompted to re-enter the credentials.
Acceptance Criteria Postcondition - The member is authenticated and given access to their account precondition - The member must have a registered account with the cooperative
Exception flow: If the system experiences server downtime during login: 1. An error message is displayed such as "The system is temporarily unavailable, please try again later."



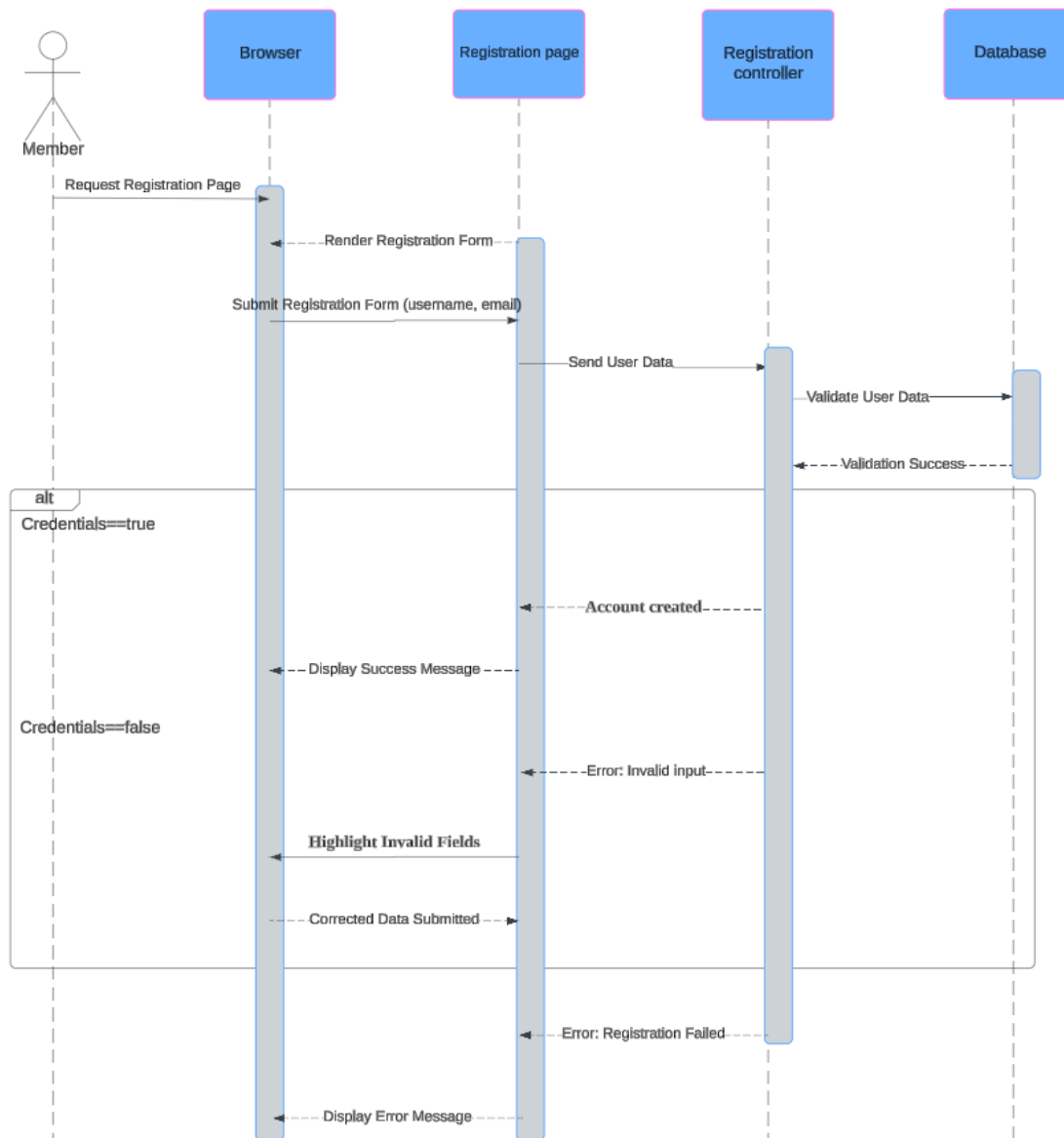


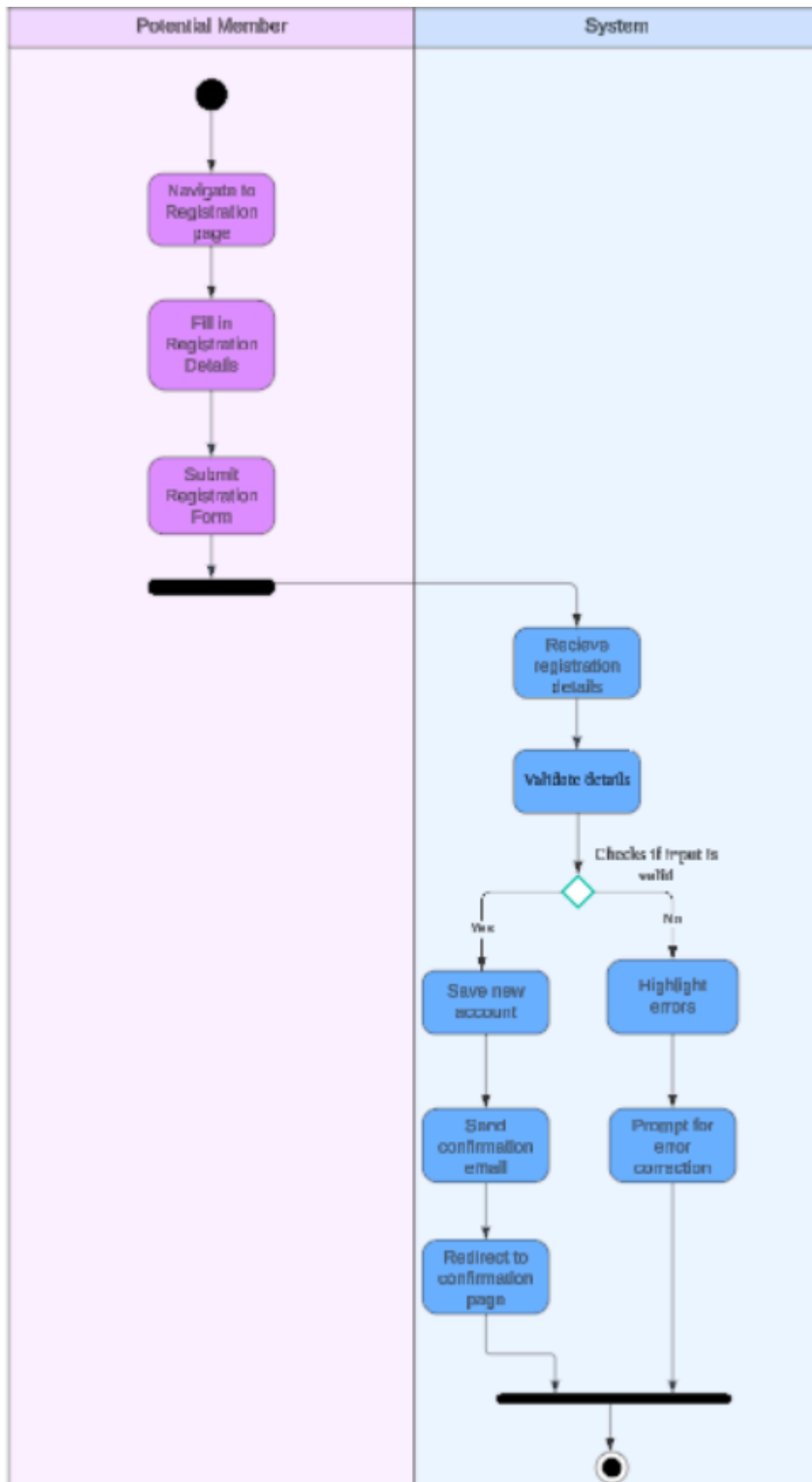
login

2.4.5 US005: User Story <Register>

Table 2.5: User Story Description for <Register>

User story: <Register>
ID: US005
User Story Description As a potential member of the cooperative I want to register for an account So that I can apply for loan
Flow of events: 1. The user navigates to the registration page 2. The user fills in the required details (username and email) given by the clerk 3. The customer submits the registration form. 4. The system validates the input and creates the account. 5. A confirmation email is sent to the customer. ...
Alternative flow n: If the customer provides invalid information: 1. The system highlights the invalid or incomplete fields with error messages. 2. The customer corrects the information and resubmits.
Acceptance Criteria Postcondition - A new account is created in the system and the customer is notified precondition - The customer must receive the designated username and password given by the clerk through email
Exception flow: If the system fails during registration submission - An error message is displayed: "Registration failed, please try again later."

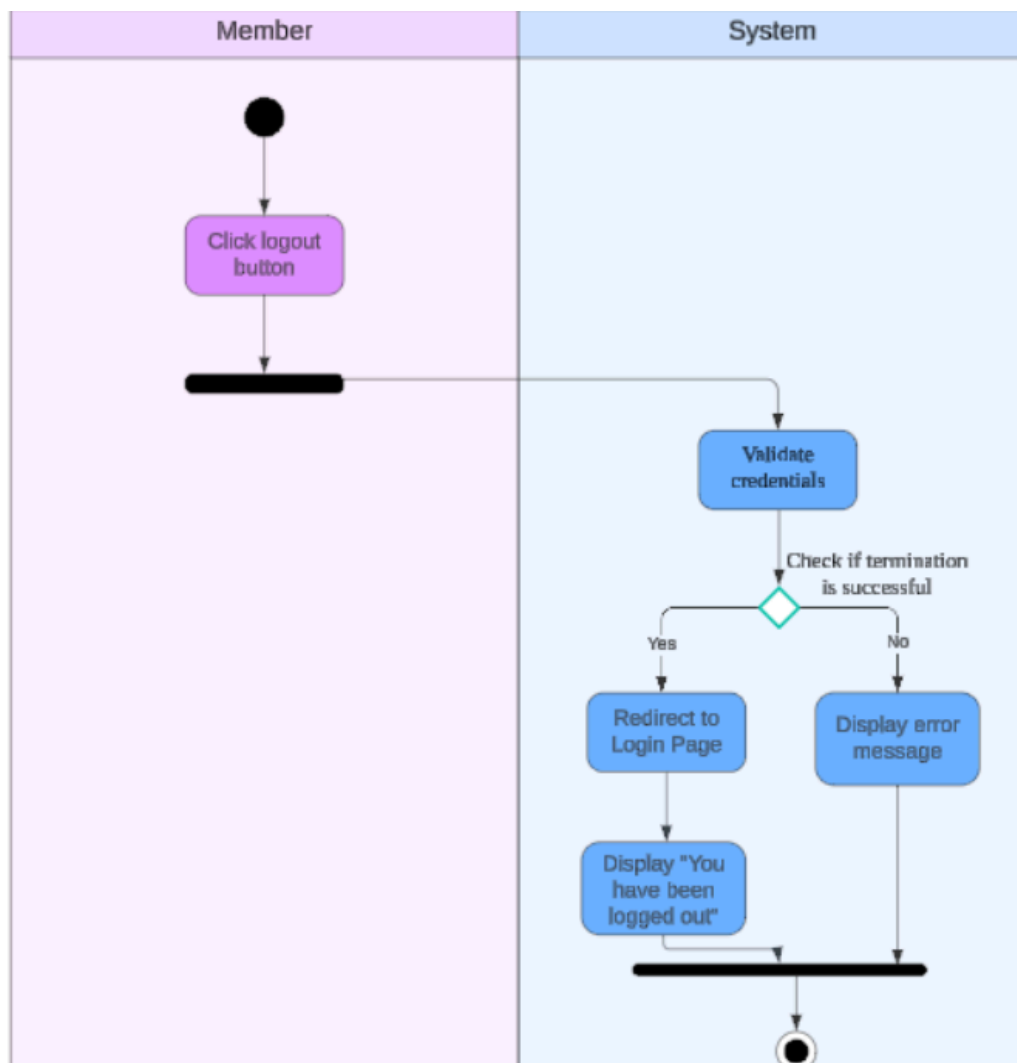
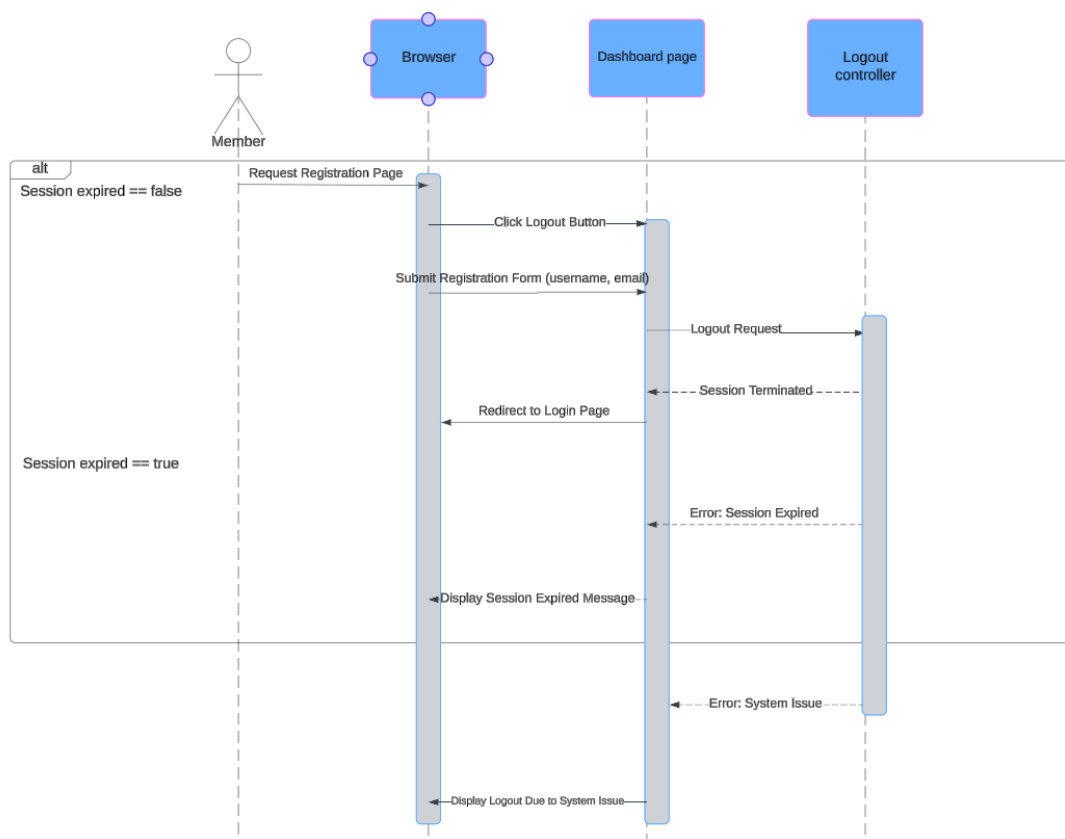




2.4.6 US006: User Story <Logout>

Table 2.6: User Story Description for <Logout>

User story: <Logout>
ID: US006
User Story Description As a member of the cooperative I want to be able to log out of the system So that I can ensure the security of my account
Flow of events: 1. The customer clicks the “Logout” button in the dashboard. 2. The system ends the session.
Alternative flow <i>n</i>: If the customer's session has already expired: - The system displays a message: Your session has expired. Please log in again.”
Acceptance Criteria Postcondition - The user session is terminated, ensuring account security precondition - The customer must be logged into the system
Exception flow: If the system encounters a server issue during logout: - The session ends, but the user is notified: “You have been logged out due to a system issue.”



2.4.7 US007: User Story <verify loan application >

Table 2.7: User Story Description for <verify loan application>

User story: <verify loan application>
ID: US007
User Story Description As a ALK I want verify loan application So that I can approve or reject loan applications
Flow of events: 1. ALK logs into the system. 2. ALK navigates to the "Loan Applications" section. 3. ALK selects a loan application from the list. 4. The system displays the loan application details. 5. ALK verifies the application and decides to approve or reject it. 6. The system updates the loan application status to "Approved" or "Rejected"
Alternative flow n: If the application has missing or invalid details: 1. The ALK is notified of the issue and can reject the application directly.
Acceptance Criteria Postcondition -The loan application status is updated to "Approved" or "Rejected" precondition -ALK must be logged into the system. ALK must have access to the loan applications section.
Exception flow: (if any in the event of error)

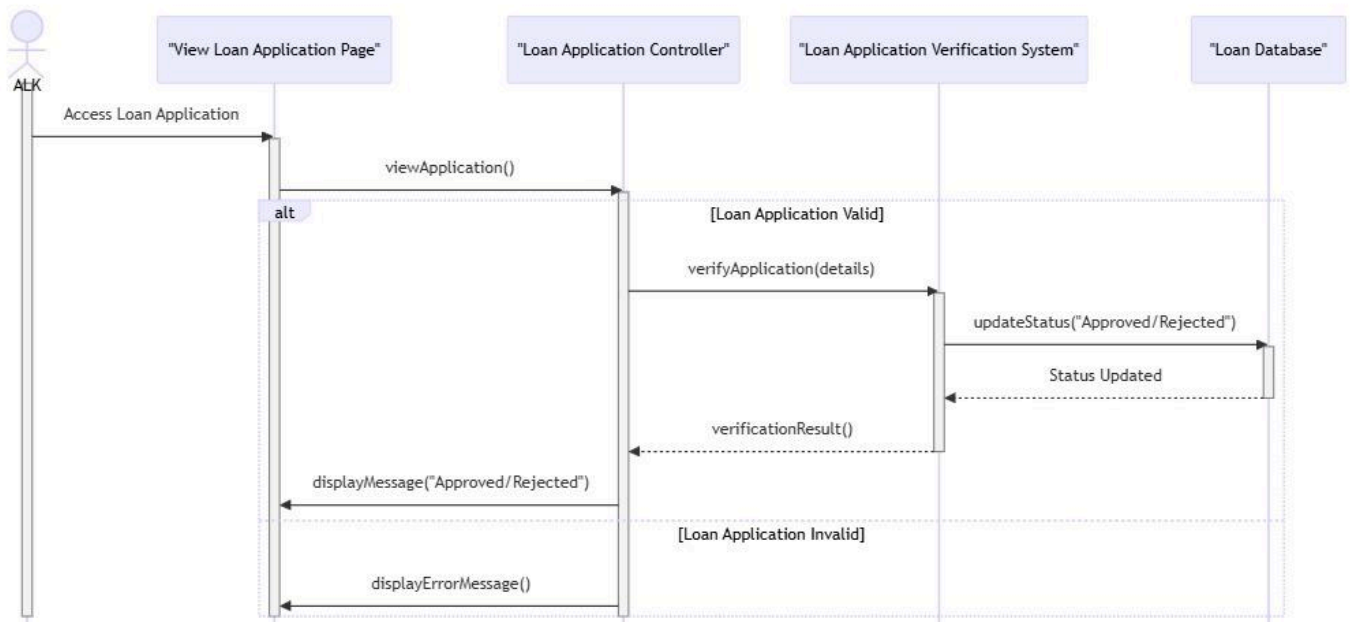


Figure 2.7.1: Sequence Diagram for <verify loan application>

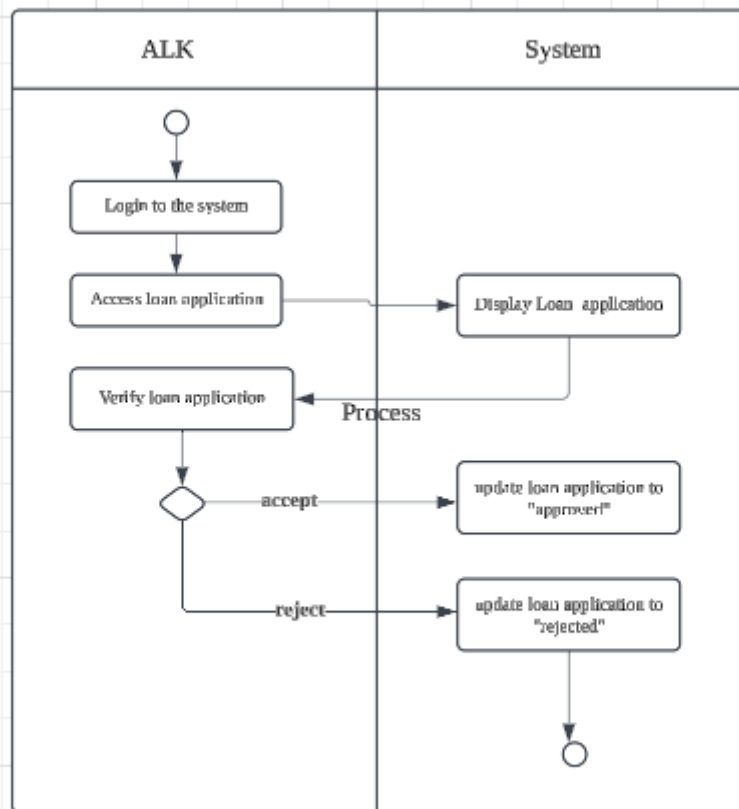


Figure 2.7.2: Activity Diagram for <verify loan application>

2.4.8 US008: User Story < view annual report >

Table 2.8: User Story Description for <View annual report>

User story: <View annual report>
ID: US008
User Story Description As a an ALK member I want view the annual report So that I can analyze the cooperative's financial performance.
Flow of events: 1. ALK members log into the system. 2. ALK members navigate to the reports section. 3. ALK members select "Annual Report" from the available reports. 4. The system displays the annual report, including financial summaries, profit and loss data, and loan statistics. 5. ALK members review the report for analysis.
Alternative flow <i>n</i>: If the ALK member needs specific data: 1. ALK member can filter the report by specific dates, loan types, or member categories. 2. System updates the report view based on the filters
Acceptance Criteria Postcondition -The ALK member successfully views the annual report. precondition -ALK member must be logged into the system and have access to financial reports. other conditions
Exception flow: 1. If the report is not available due to a system error, the ALK member receives a notification to retry later

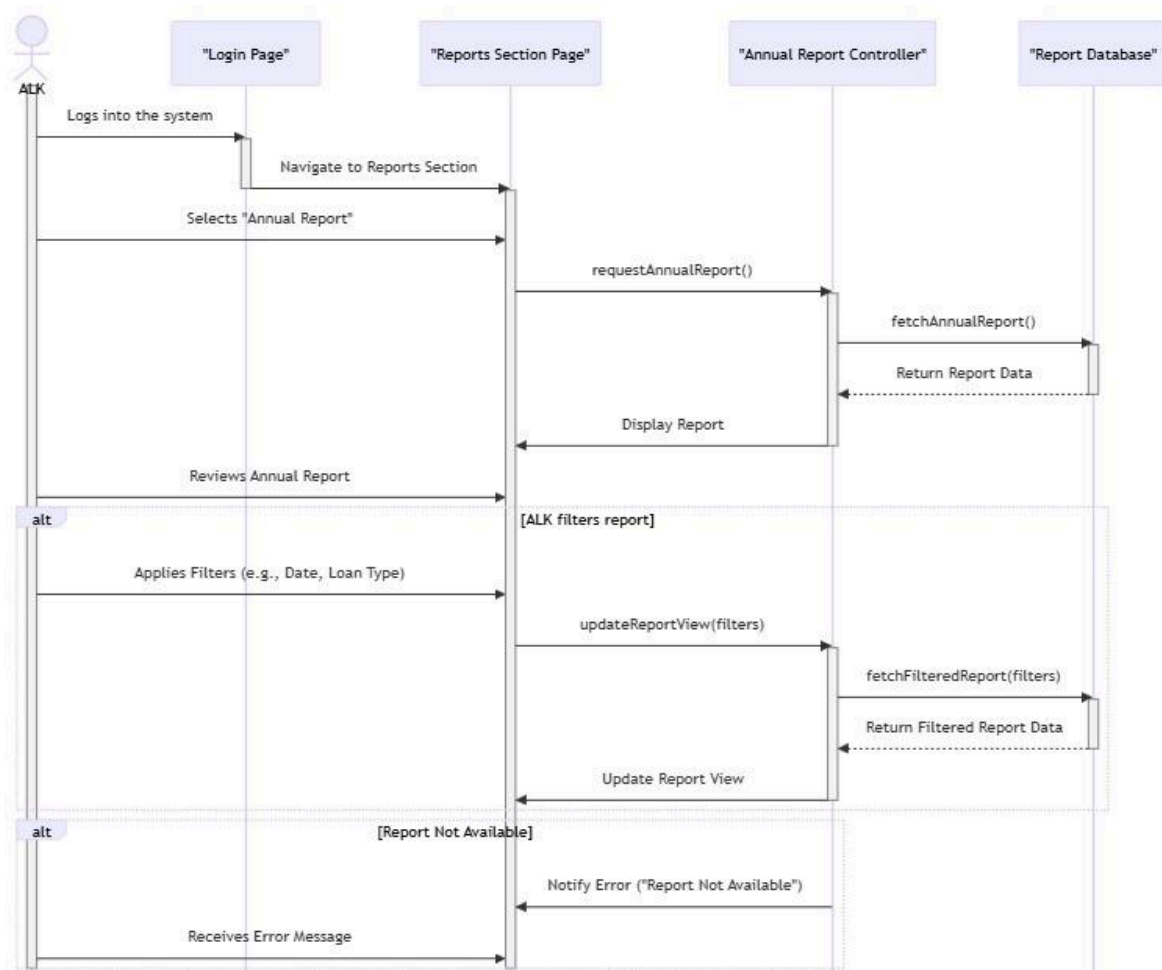


Figure 2.8.1: Sequence Diagram for <view annual report>

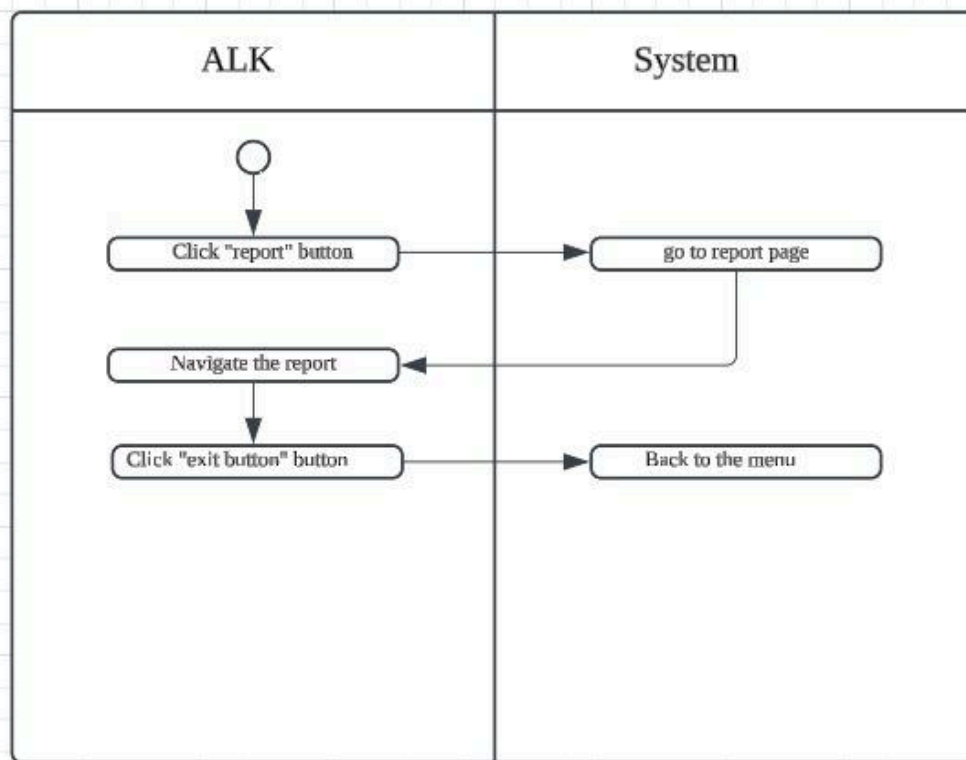


Figure 2.8.2: Activity Diagram for <view annual report>

2.4.9 US009: User Story <create an account>

Table 2.9: User Story Description for <Create an account>

User story: <Create an account>
ID: US009
User Story Description As a clerk of the KADA cooperative I want to create an account for the staff So that I can monitor who is the one requested to be a KADA cooperative member
Flow of events: 1. the clerk will check the list of the new register member 2. the clerk will make sure the that the one who is register is from KADA 3. the clerk will click the button 'accept' and it will lead to the create account page 4. the clerk will create an account for them by using their email, staff id , name. 5. the clerk will click the button 'submit' to indicate they are officially a member and it will notify and give a account username and password to the applicant by email
Alternative flow <i>n</i>: 1. the new register member are not from KADA 1.1. the clerk check the staff id and it wrong 1.2. the clerk will push the button 'reject' to clear up the list of new register member and it will automatically send an email to the applicant to indicate that the application has been decline
Acceptance Criteria Postcondition - the account has been successfully create precondition - the clerk must verify the staff id whether it's true or not
Exception flow:

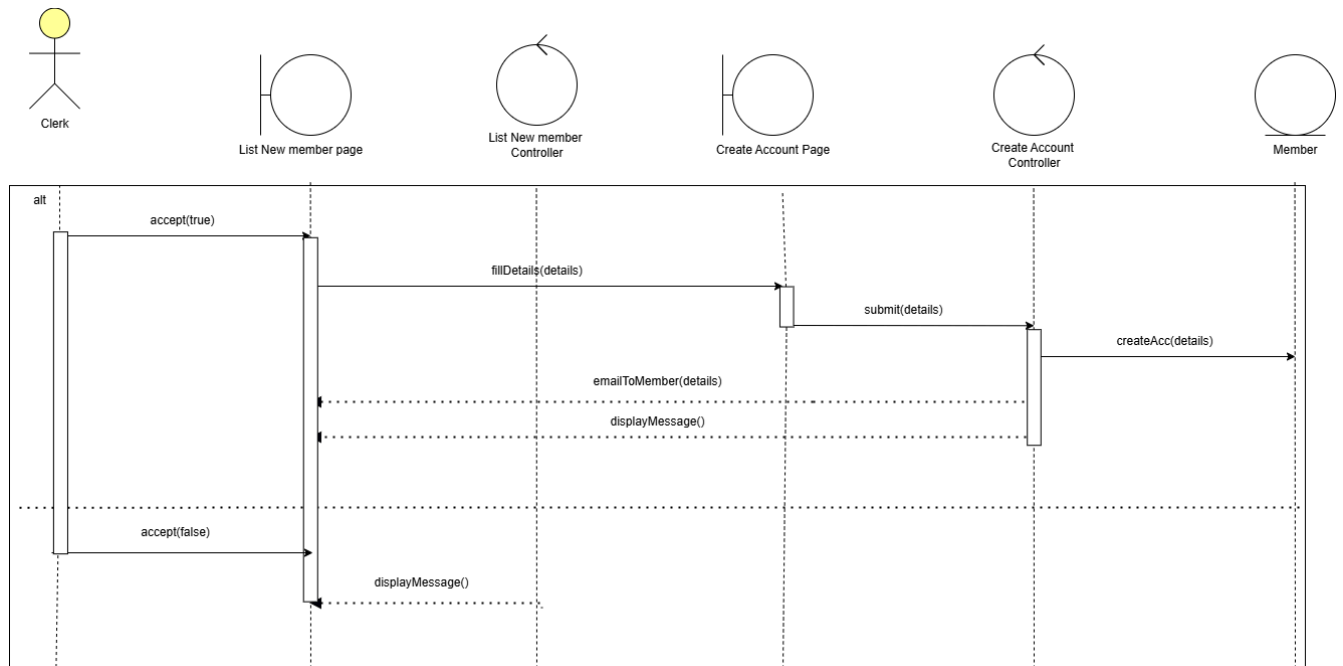


Figure 2.9.1: Sequence Diagram for <create an account>

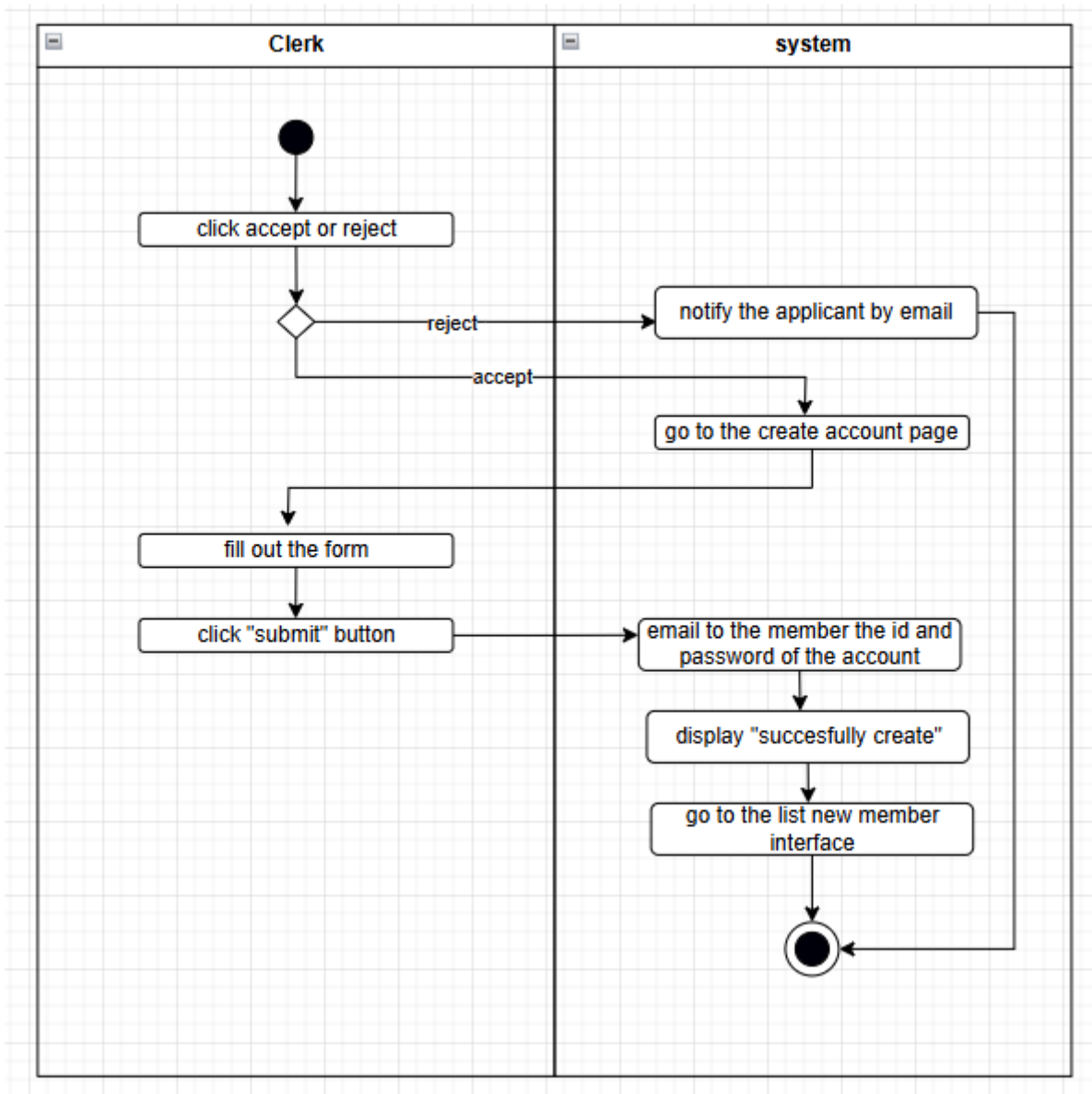


Figure 2.9.2: Activity Diagram for <create an account>

2.4.10 US010: User Story <manage list of member>

Table 2.10: User Story Description for <Manage list of member>

User story: <Manage list of member>
ID: US010
User Story Description As a clerk of the KADA cooperative I want to be able to manage the list of the member that has been successfully register as a member of KADA cooperative So that I can update, delete, view their personal information and ect
Flow of events: 1. the clerk can monitor overall information of the members
Alternative flow <i>n</i>: 1. members that already retired or quit 1.1. the clerk will push button 'delete' to delete the information of the members
Acceptance Criteria Postcondition - the clerk must login as a clerk with clerk ID and password in login section precondition - the clerk can monitor overall information of the cooperative members
Exception flow:

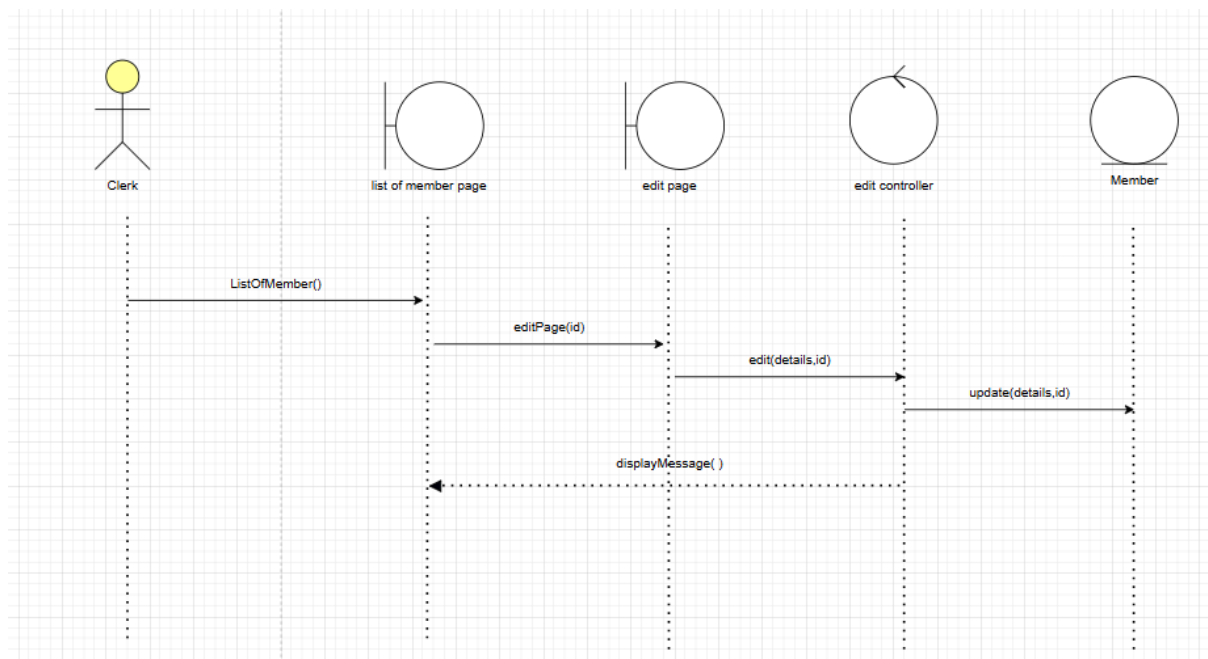


Figure 2.10.1: Sequence Diagram for <manage list of member (edit)>

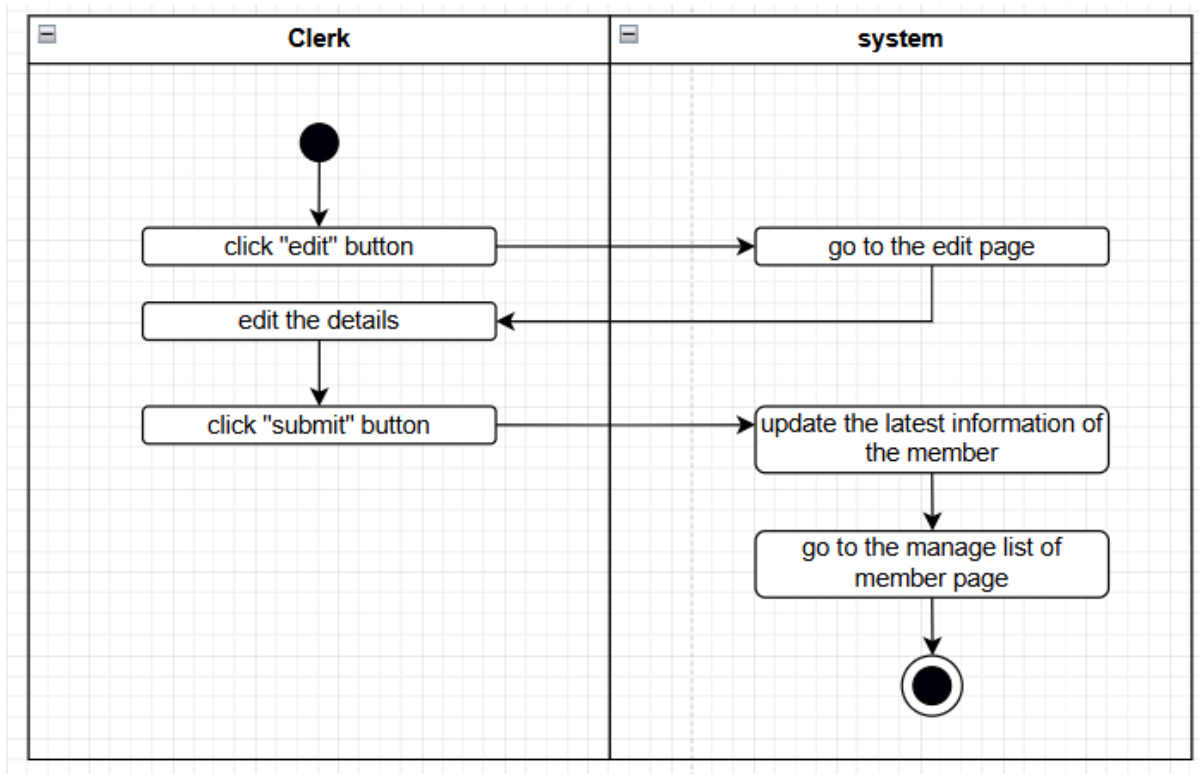


Figure 2.10.2: Activity Diagram for <manage list of member (edit)>

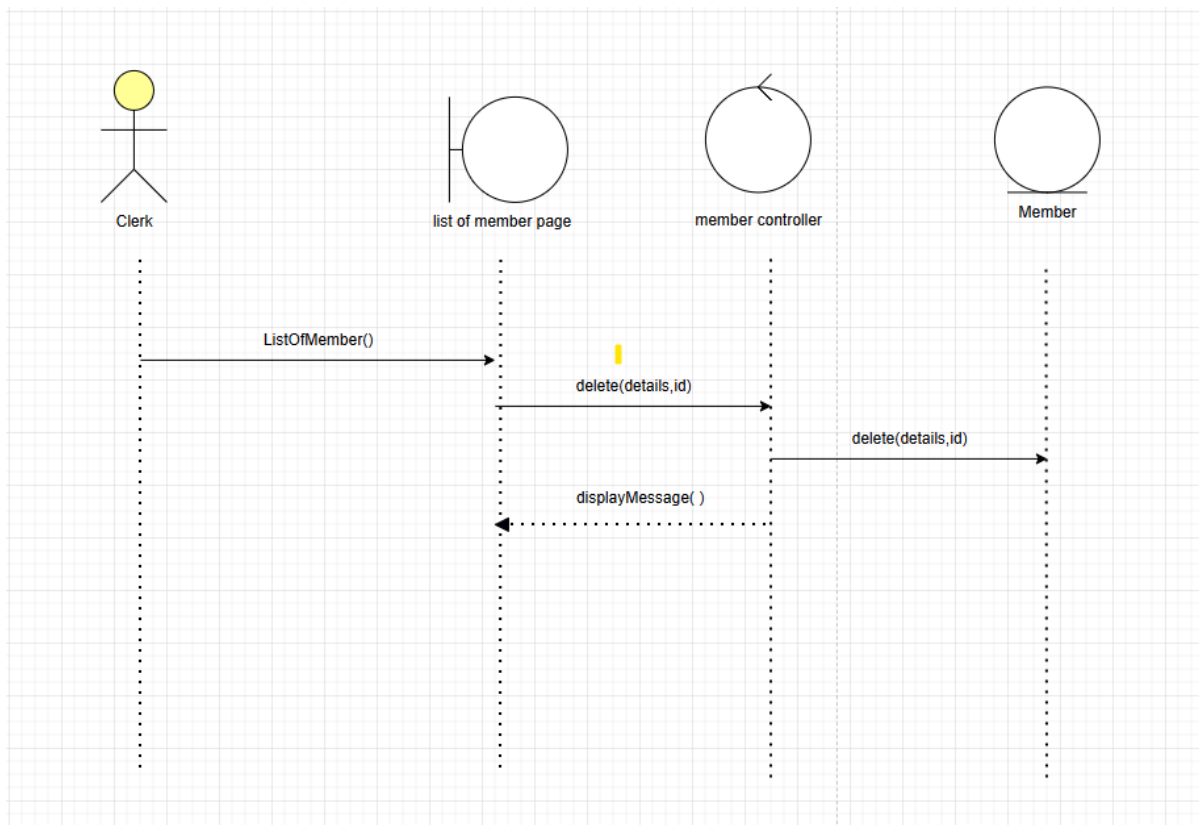


Figure 2.10.3: Sequence Diagram for <manage list of member (delete)>

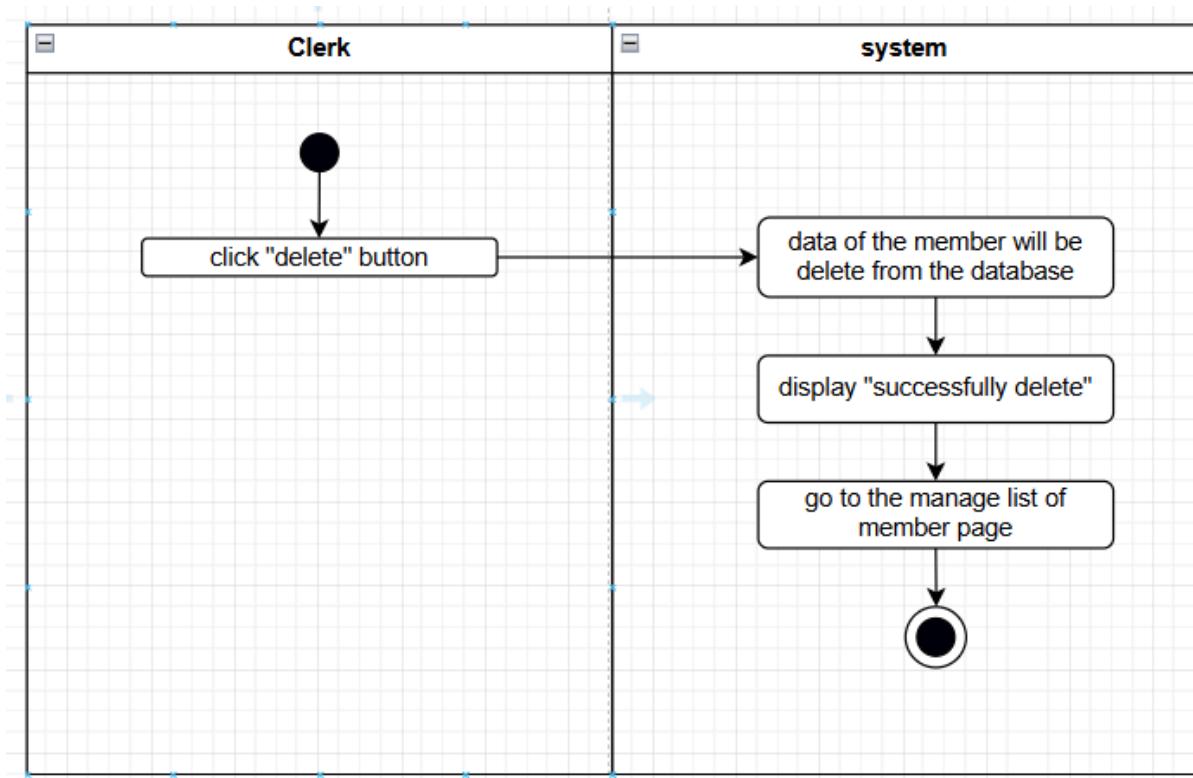


Figure 2.10.4: Activity Diagram for <manage list of member (delete)>

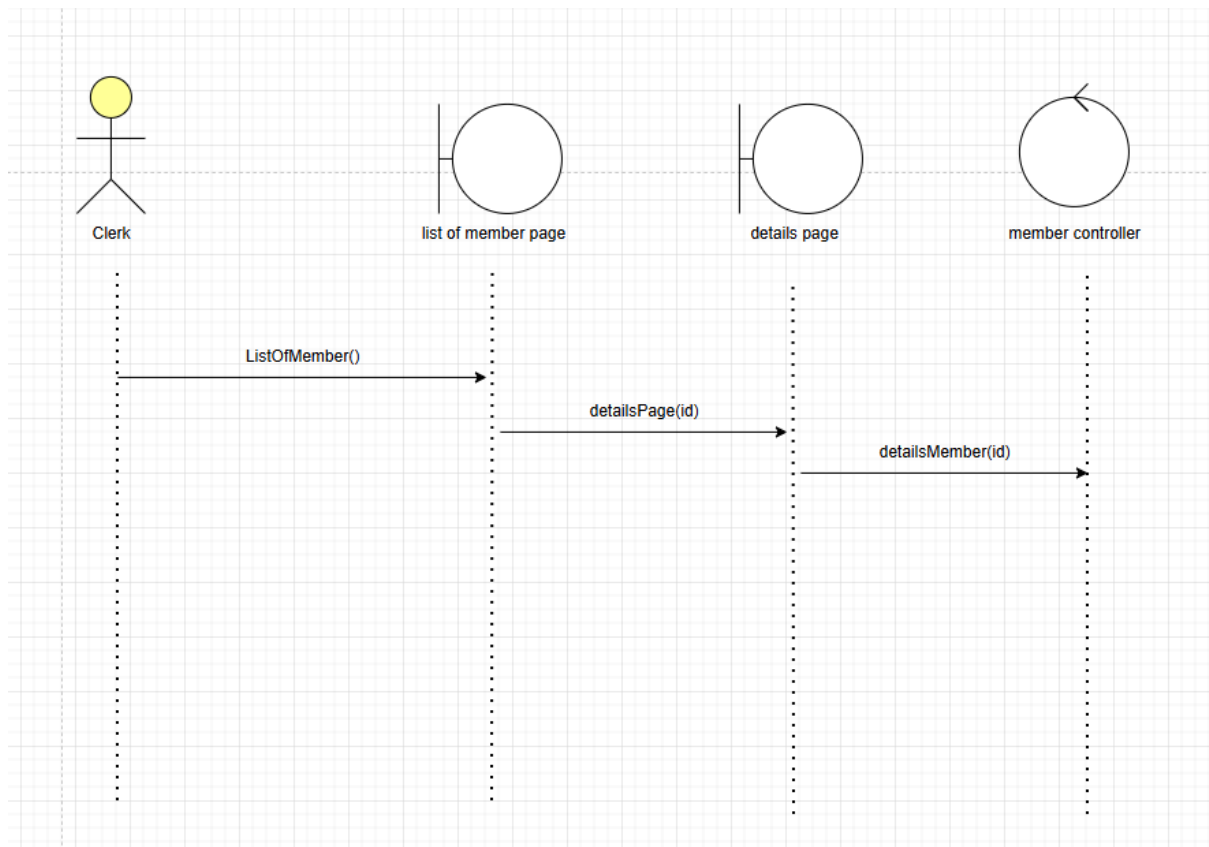


Figure 2.10.5: Sequence Diagram for <manage list of member (view details)>

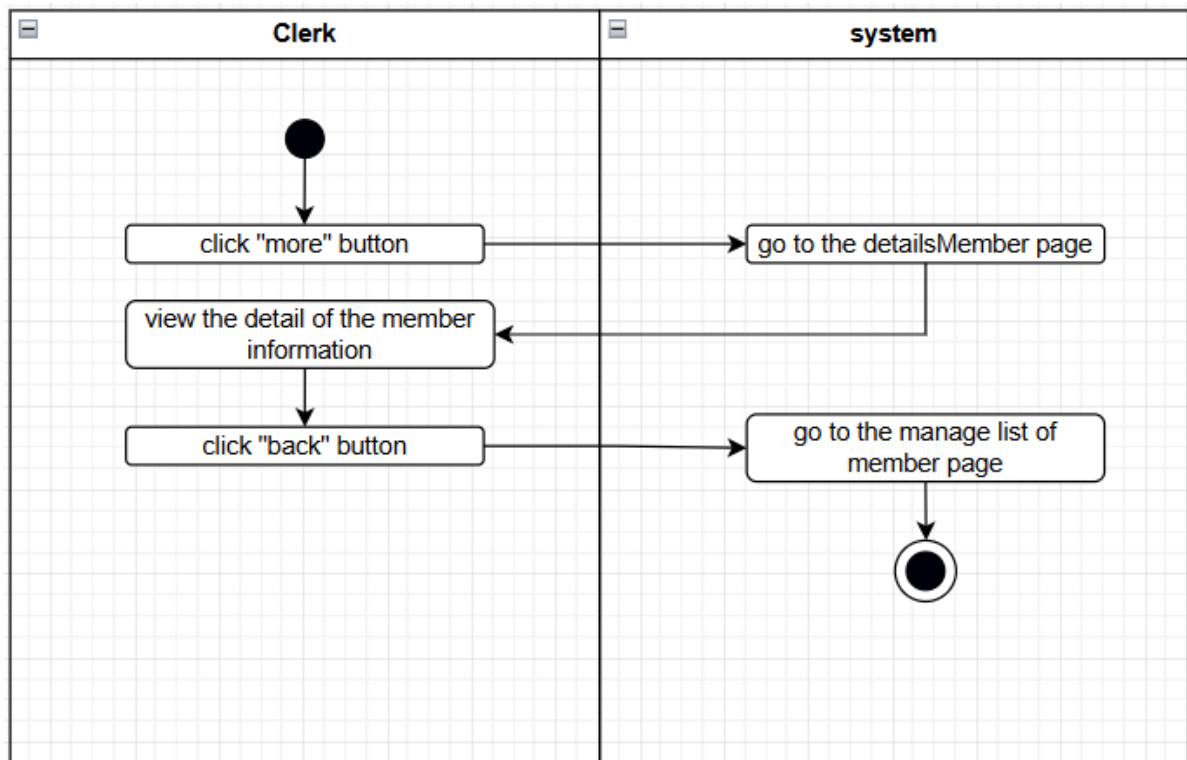


Figure 2.10.6: Activity Diagram for <manage list of member (view details)>

2.4.11 US011: User Story <manage loan application>

Table 2.11: User Story Description for <Manage loan application>

User story: <Manage loan application>
ID: US011
User Story Description As a clerk of the KADA cooperative I want to be able to manage the loan applicant that been request by the cooperative member So that I can check whether the applicant are qualify or not to apply the loan
Flow of events: 1. the clerk will go to the manage loan application 2. the clerk will go through all of the loan applicant details one by one by push the button “more” to see the overall information 3. the clerk than will tick at the button “red” or “green” that indicate whether the loan applicant are qualify or not
Alternative flow <i>n</i>:
Acceptance Criteria Postcondition - the clerk must login as a clerk with clerk ID and password in login section precondition - the clerk can monitor overall information of the loan applicant
Exception flow:

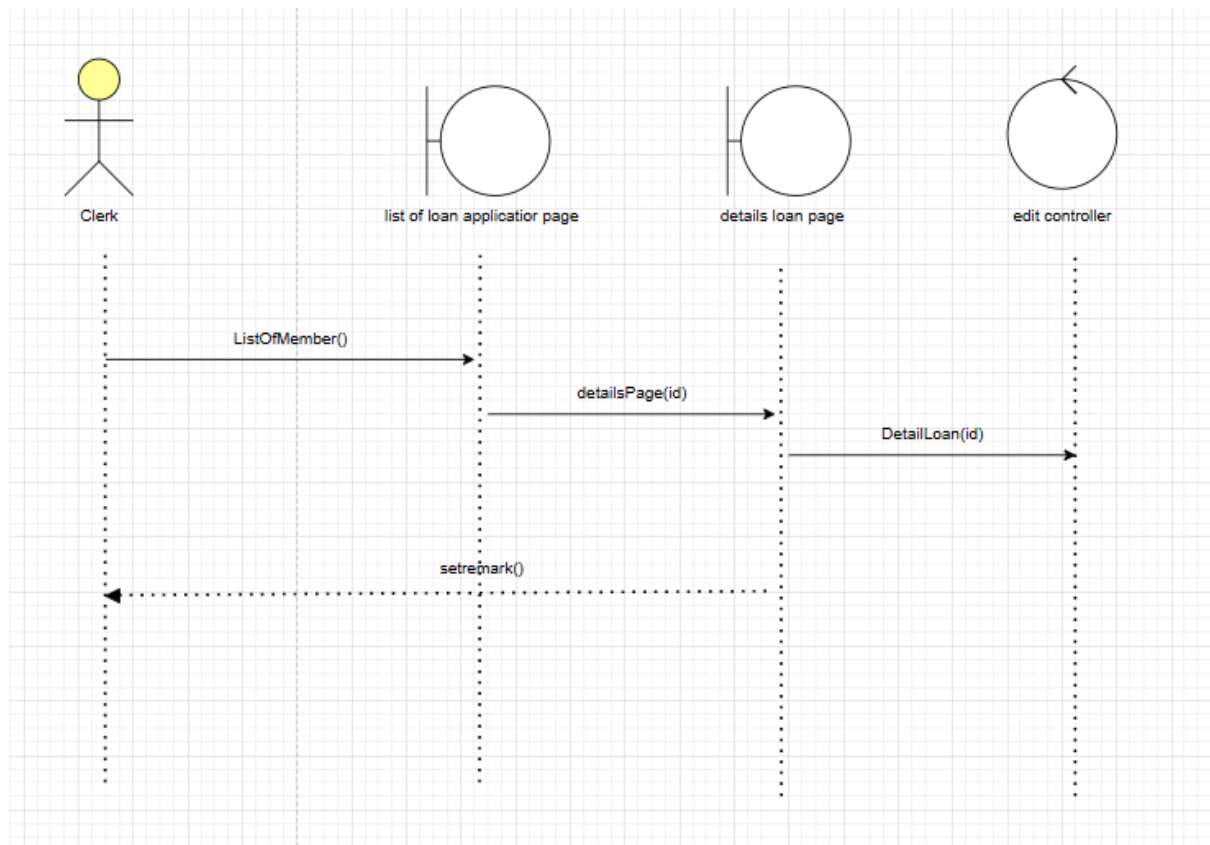


Figure 2.11.1: Sequence Diagram for <manage loan application>

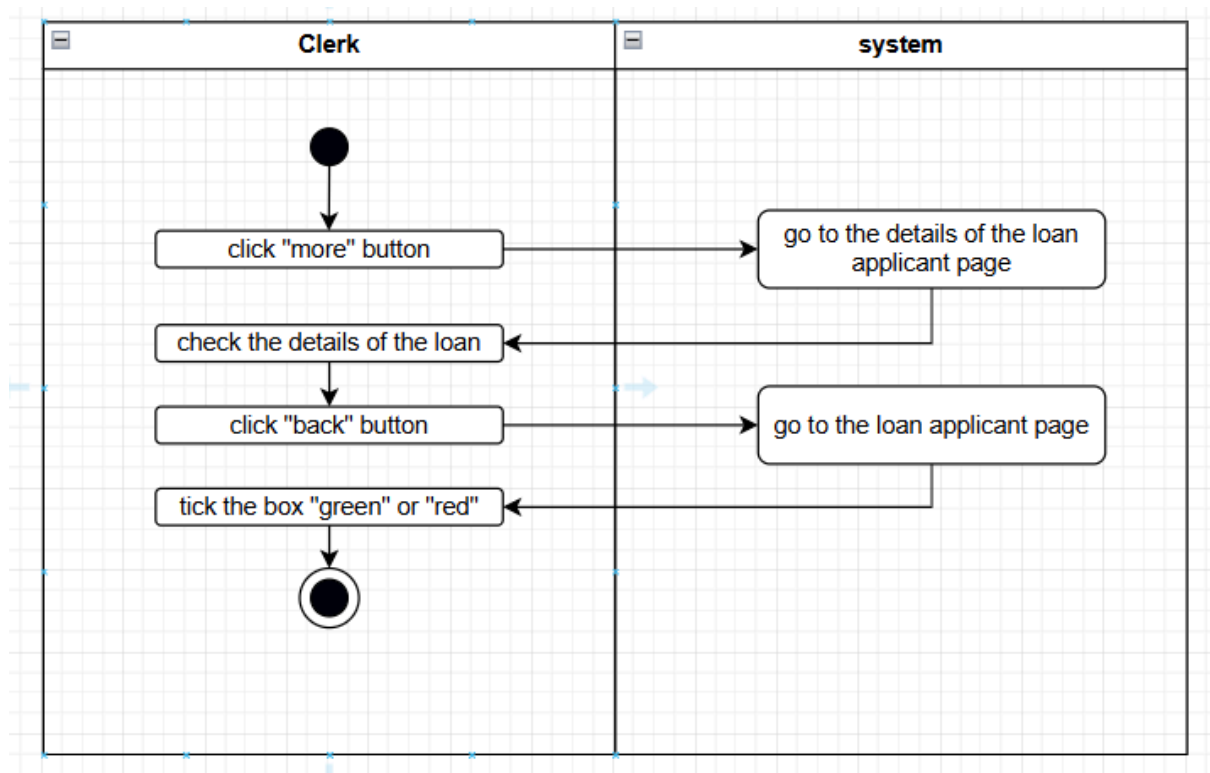


Figure 2.11.2: Activity Diagram for <manage loan application>

2.4.12 US012: User Story < *calculate loan payment* >

Table 2.12: User Story Description for <Calculate Loan Payment>

User story: <Calculate loan payment>
ID: US012
User Story Description As a member of KADA I want to be able to calculate my loan payment before applying it So that I know how much money I will get, the amount of repayment. and the duration of it.
Flow of events: 1. Member log in into the cooperative system 2. Member navigate to the loan calculator section 3. Member enter the loan type, loan amount, and the duration of repayment 4. Member click the “calculate” button 5. The system will calculate the loan and display the total repayment amount and the monthly repayment within the time duration provided. 6. Member reviews the calculated information
Alternative flow <i>n</i>:
Acceptance Criteria Postcondition: - Members acknowledge the loan information such as the repayment amount, and the duration of paying back. precondition: - Must be a member of KADA and log in
Exception flow:

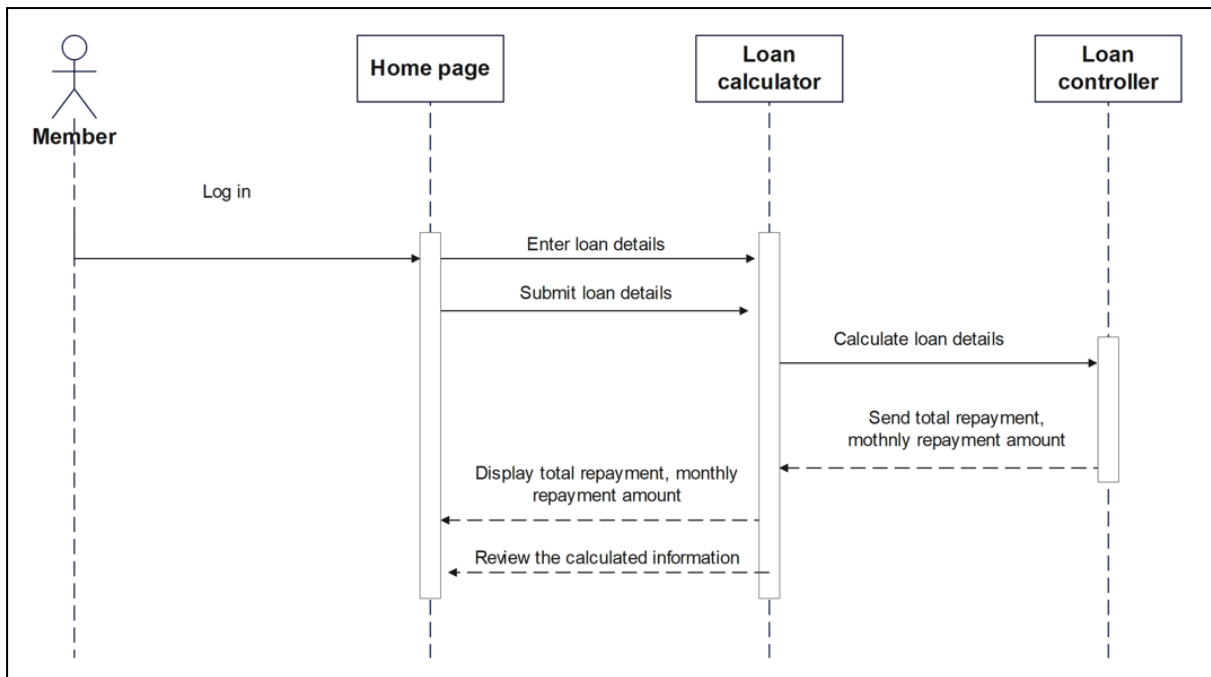


Figure 2.12.1: Sequence Diagram for <Calculate Loan Payment>

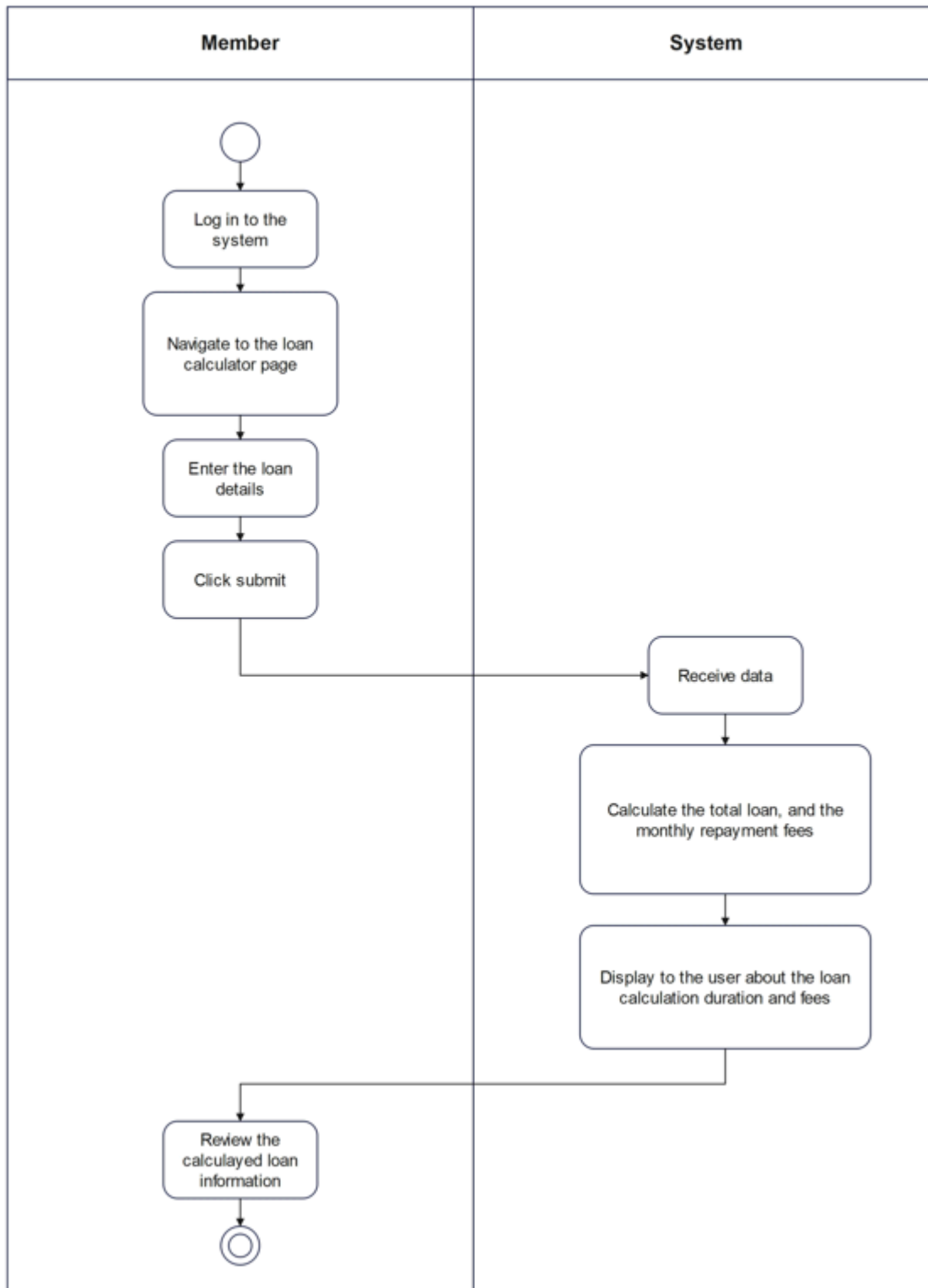


Figure 2.12.2: Activity Diagram for <Calculate Loan Payment>

2.5 Performance and Other Requirements

Non-functional requirements are the requirements that describe how the system should behave or operate, rather than what it should do. They address the characteristics or qualities of the system, such as its usability, reliability, performance, security, maintainability, and compatibility. Non-functional requirements are important to write in the SD because they help ensure that the system meets the stakeholders' needs and expectations. The ISO/IEC/IEEE 29148 standard suggests that non-functional requirements should be specified under three main categories: Software System Attributes, Performance and Other Requirements.

Example of Performance is as follows:

Performance requirements define the system's capability to respond to user requests and handle data in a timely manner. These requirements include the following:

- *Response Time: The time it takes for the system to respond to a user request.*
- *Throughput: The number of requests the system can handle in a given period of time.*
- *Capacity: The maximum number of users or amount of data that the system can handle.*
- *Availability: The percentage of time the system is operational and accessible to users.*

Performance requirements

Response time:

The system must provide timely responses to user interactions, such as submitting feedback, retrieving reports, and executing tasks. This ensures a seamless and efficient user experience, minimizing any frustration caused by delays.

Throughput:

The system must support a high number of simultaneous user requests, maintaining effective processing speeds and ensuring minimal wait times even during peak usage periods. This capability is critical to accommodate multiple users without performance degradation.

Capacity:

The system must efficiently manage and store a substantial volume of feedback submissions and related data. It should be designed to scale and support the entire university's student population, courses, and associated information without compromising performance.

Availability:

The system should maintain a consistently high uptime rate, ensuring it remains accessible and functional for users. It must minimize downtime and disruptions to provide reliable service at all times.

Other requirements

Security:

The system must implement comprehensive security measures, such as secure user authentication, robust data encryption, and strict access controls. These features are essential to safeguard student data and prevent unauthorized access or potential data breaches.

Safety:

The system must operate in a manner that prioritizes user safety and security, ensuring it does not present any risks or harm to users or the broader university environment. This includes avoiding technical vulnerabilities and ensuring system reliability.

Legal and regulatory:

The system must adhere to all applicable laws, regulations, and standards related to data privacy, confidentiality, and information security. Compliance ensures legal integrity and builds trust among users and stakeholders.

Maintainability:

The system is designed to be easily modified or enhanced due to the implementation of good coding practices. This includes thorough comments and the use of meaningful, descriptive variable names, ensuring maintainability and readability for developers.

Usability:

The system shall provide an intuitive, well-organized, and user-friendly interface to facilitate ease of use, enabling users to navigate and perform tasks without confusion.

Example of Software System Attributes are as follows:

Software system attributes define the overall qualities or characteristics of the software. These attributes are the foundation on which the software is built, and they include the following:

- *Usability: The ease with which users can interact with the system.*
- *Reliability: The ability of the system to perform its functions correctly and consistently.*
- *Maintainability: The ease with which the system can be modified, repaired, and enhanced.*
- *Portability: The ability of the system to run on different hardware and software platforms.*
- *Compatibility: The ability of the system to work with other systems and components.*
- *Security: The protection of the system and its data from unauthorized access and malicious attacks.*
- *Safety: The ability of the system to operate safely without causing harm to users or the environment.*
- *Legal and Regulatory: Compliance with laws, regulations, and standards.*
- *Environmental: The impact of the system on the environment.*

2.6 Design Constraints

1. Environmental Constraints

The system needs to function well in environments with varying temperatures, ranging from freezing cold at 0°C to warm conditions of up to 40°C. This ensures it remains reliable even in locations without temperature-controlled server rooms. Additionally, the interface should be clear and usable whether users are in dimly lit areas or working under bright sunlight.

2. Hardware Constraints

The software should be lightweight enough to run on devices with basic specifications, like 4GB of RAM and dual-core processors, making it accessible to a wider range of users. For data storage, the database must have a minimum capacity of 1TB to accommodate the cooperative's expanding records, including member data and financial transactions.

3. Security Constraints

To protect sensitive information, such as members' personal and financial details, the system must use advanced encryption standards like AES-256. For added safety, key users like ALK members and clerks should use multi-factor authentication when accessing the platform. Moreover, the system should comply with data privacy regulations, including GDPR and Malaysia's Personal Data Protection Act (PDPA).

4. Compatibility Constraints

The platform should work seamlessly across popular web browsers like Chrome, Firefox, and Edge. It also needs to support both iOS and Android devices, ensuring accessibility for users on mobile. Integration with existing tools used by KADA and third-party services like email or SMS notification systems is essential to streamline operations.

5. Performance Constraints

The system should handle up to 500 users online at the same time, with responses delivered within 2 seconds on average. For bulk processing, such as reviewing a thousand loan applications at once, the process shouldn't take more than 10 minutes. Finally, to meet the cooperative's operational needs, the system must be highly reliable, maintaining an uptime of 99.9%.